Card column reference

All **711** columns across the five cards, grouped by card and then by block, with blocks ordered by **measured importance**. Each entry leads with what the column *is*; the caveats fold underneath.

Generated 2026-08-20 by `analyses/ops/gen_column_reference.py` from `card_glossary.tsv` and `card_registry.tsv`. **Do not hand-edit.**

Vintage mix â€” read before quoting a fitted column

The delivered edge card is **16 days newer** than the base it layers over. Its **84 base-owned columns** (`coupling_`, `beta_`, `dose_`, `share_`, `rank_`) carry the older vintage; the other 102 are current.

The one rule behind this layout

The rung is a property of (CARD, COLUMN) â€” never of the column name. `beta` is edge-rung on the edge card and family-rung on the gene_family card: same estimator, different unit. That is why this is organised card-first, and why one name can appear twice with different text.

How blocks are ordered

By three measured signals, not by size: how often a block's columns are named in the **discovery registry** (what the findings rest on, weighted highest), how many **modules** read them, and whether the block is **base-owned** (fitted) rather than an annotation join. Names under 6 characters count only backtick-quoted registry hits â€” otherwise `n` and `z` match ordinary prose.

One correction the measurement needed: mention-frequency is not importance â€” a *settled* block gets discussed less precisely because nobody argues about it. `adm_` scored 27th of 44 on the raw signals while MH-216 calls `adm_has_site` the single most load-bearing conditioning variable, on 4 registry mentions. Five blocks therefore carry a **curated floor**, each citing the finding that justifies it (see `CURATED_FLOOR`). That is the only hand-set input here.

Cards

- `edge â€” 186 columns, 47 blocks`
- `gene â€” 135 columns, 29 blocks`
- `gene_family â€” 61 columns, 12 blocks`
- `arm â€” 296 columns, 44 blocks`
- `seed_family â€” 33 columns, 2 blocks`

realization/edge_card.tsv

edge

One row per one (miRNA arm, gene) pair. The widest card and the one most claims are made from. 84 of its columns come from the BASE card and 102 from the annotation layer â€” check the vintage banner before quoting a fitted one.

186 columns **5,649** rows **47** blocks median coverage **72%**

Contents â€" 47 blocks, most important first

(BARE) Â· 6

core â€" the columns findings rest on

6 columns Â· **spans edge, key** â€" this prefix groups columns living on DIFFERENT units Â· coverage **96%â€"100%** (median 100%)
2× identifier Â· 2× real â‰¥ 0 Â· 1× boolean Â· 1× real, signed

arm
KEY IDENTIFIER 100%

KEY â€" mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through `_arm_key()`, which gates against the canonical universe and LOGS drops rather than silently dropping. AUTHORED

example: `CDH1 / miR-25-3p/32-5p/92-3p/363-3p/367-3p â†' hsa-miR-25-3p`

beta
EDGE REAL Â‰¥ 0 100%

EDGE-rung coupling coefficient â€" `readouts.run(level='arm')`, the same-seed collapse REMOVED. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.2203`

gene

KEY IDENTIFIER 100%

KEY â€" HGNC gene symbol. AUTHORED

example: `hsa-miR-21-5p / miR-21-5p/590-5p â†' ACAT1`

---

identified

EDGE BOOLEAN 100%

`|z| > 2` on the UNCALIBRATED posterior SD, where `z = beta / beta_sd` (MH-83 records precision 0.86 / recall 0.89 against a held-out standard). **True on 24.8% of edges.** AUTHORED

values: `False (4,269) Â· True (1,375)`

example: `CDH1 / hsa-miR-25-3p â†' True`

---

identity

EDGE REAL, SIGNED 96%

Shapley/LMG share of the gene's R^2 credited to this edge, with bagged-NNLS weights. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.3367`

---

z

EDGE REAL â‰¥ 0 100%

`beta / beta_sd` for the EDGE-level fit (`readouts.run(level='arm')`, the whole gene's arms as separate predictors), on the UNCALIBRATED posterior SD. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 4.542`

SEED_ Â· 1

core â€" the columns findings rest on

1 columns Â· rung **family** Â· coverage **100%â€"100%** (median 100%)
1× identifier

seed_family

FAMILY IDENTIFIER 100%

KEY â€" the seed family itself, one row per family. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' miR-25-3p/32-5p/92-3p/363-3p/367-3p`

COUPLING_ Â· 13

core â€" the columns findings rest on

13 columns Â· **spans edge, family** â€" this prefix groups columns living on DIFFERENT units Â· coverage **3%â€"100%** (median 62%)
5× signed âˆ'1â€¦1 Â· 5× p-value Â· 3× real, signed

coupling_fam

FAMILY SIGNED âˆ'1â€¦1 100%

Family-level coupling for the edge's (gene, seed_family) cell. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' -0.2577`

---

coupling_hly

EDGE SIGNED âˆ'1â€¦1 56%

Partial-Spearman coupling of arm dose against target mRNA in the GTEx-HEALTHY state, conditioned on C (composition + target CN + proliferation). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.007`

---

coupling_hly_surrogate

EDGE SIGNED âˆ'1â€¦1 3%

Healthy coupling via a same-seed SURROGATE arm, used where the arm itself has no GTEx signal. AUTHORED

example: `CDH1 / hsa-miR-30a-5p â†' -0.037`

---

coupling_nat

EDGE SIGNED âˆ'1â€¦1 62%

Partial-Spearman coupling of arm dose against target mRNA in matched NORMAL (NAT), conditioned on C (composition + target CN + proliferation). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' -0.116`

---

coupling_p_hly

EDGE P-VALUE 56%

The site-free-null p for that coupling in the GTEx-HEALTHY state, conditioned on C (composition + target CN + proliferation). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.6225`

coupling_p_hly_resolved
EDGE P-VALUE 59%

The site-free-null p for that coupling in the GTEx-HEALTHY state, conditioned on C (composition + target CN + proliferation). AUTHORED
example: `CDH1 / hsa-miR-25-3p â†' 0.6225`

---

coupling_p_hly_surrogate
EDGE P-VALUE 3%

The site-free-null p for that coupling in the GTEx-HEALTHY state, conditioned on C (composition + target CN + proliferation). AUTHORED
example: `CDH1 / hsa-miR-30a-5p â†' 0.4457`

---

coupling_p_nat
EDGE P-VALUE 62%

The site-free-null p for that coupling in matched NORMAL (NAT), conditioned on C (composition + target CN + proliferation). AUTHORED
example: `CDH1 / hsa-miR-25-3p â†' 0.1658`

---

coupling_p_tum
EDGE P-VALUE 69%

p against the CALIBRATED site-free null (not a theoretical t-null). Median 0.375; <0.05 on 16.2%. AUTHORED
example: `CDH1 / hsa-miR-25-3p â†' 0.0012`

---

coupling_tum
EDGE SIGNED Âˆ'1â€¦1 72%

Partial-Spearman coupling of arm dose against target mRNA in TUMOUR, conditioned on C (composition + target CN + proliferation). AUTHORED
example: `CDH1 / hsa-miR-25-3p â†' -0.271`

---

coupling_z_hly
EDGE REAL, SIGNED 59%

That coupling expressed in NULL SDs â€" the honest effect size in the GTEx-HEALTHY state, conditioned on C (composition + target CN + proliferation). AUTHORED
example: `CDH1 / hsa-miR-25-3p â†' 0.31`

---

coupling_z_nat
EDGE REAL, SIGNED 62%

That coupling expressed in NULL SDs â€" the honest effect size in matched NORMAL (NAT), conditioned on C (composition + target CN + proliferation). AUTHORED
example: `CDH1 / hsa-miR-25-3p â†' -0.97`

---

coupling_z_tum
EDGE REAL, SIGNED 69%

The tumour coupling expressed in NULL SDs â€" the CALIBRATED effect size, and the scale to quote. Median **âˆ'0.32**; |z|>2 on 15.5%, |z|>3 on 6.4%. AUTHORED
example: `CDH1 / hsa-miR-25-3p â†' -3.04`

ADM_ Â· 4

core â€" the columns findings rest on

> **4** columns Â· rung **edge** Â· per-edge FLAGS; the arm card carries their rollup Â· coverage **87%â€"87%** (median 87%)
> 4× boolean

adm_admissible
EDGE BOOLEAN 87%

The admissibility verdict: site + evidence + expression. AUTHORED
values: `True (2,981) Â· False (1,956)`
example: `CDH1 / hsa-miR-25-3p â†' True`

---

adm_expr_gtex
EDGE BOOLEAN 87%

ADMISSIBILITY â€" could this edge repress in breast AT ALL? `has_site` (seed match in the 3'UTR) AND `expressed` (arm present in TCGA tumour OR GTEx healthy breast). AUTHORED
values: `False (3,927) Â· True (1,010)`
example: `CDH1 / hsa-miR-25-3p â†' True`

---

adm_expr_tcga
EDGE BOOLEAN 87%

Is the arm expressed above floor in TCGA? AUTHORED
values: `True (3,154) Â· False (1,783)`
example: `CDH1 / hsa-miR-25-3p â†' True`

---

adm_has_site
EDGE BOOLEAN 87%

Does the arm have a seed match in this target's 3'UTR. AUTHORED
values: `True (4,436) Â· False (501)`
example: `CDH1 / hsa-miR-25-3p â†' True`

ARM_ Â· 13

core â€” the columns findings rest on

13 columns Â· **spans arm, arm-in-family** â€” this prefix groups columns living on DIFFERENT units Â· coverage **20%â€“100%** (median 69%)
4× real, signed Â· 3× fraction 0â€“1 Â· 2× categorical Â· 2× real â‰¥ 0 Â· 1× percent Â· 1× boolean

arm_credit_share
ARM-IN-FAMILY · FRACTION 0Â€“1 · 20%

ââ **NOT 'which arm is the real regulator' â€” that reading is exactly why it was renamed away from `identity_arm` (MH-188).** It is an exact Shapley over the cell's arm columns on RÂ²(arms, y|C); the shares sum to **1.0000 on 100% of cells** (verified), and it is defined ONLY on multi-arm cells â€” 1,147 edges, 20.3%. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 0.7094

---

arm_detection
ARM · FRACTION 0Â€“1 · 95%

Fraction of patients in which this ARM is measured at all (`readouts._detection`). AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 1

---

arm_id_status
ARM-IN-FAMILY · CATEGORICAL · 100%

How the arm's identity within its cell was resolved: `resolved_by_dose` 2,344 Â· `singleton` 2,260 Â· `unvalidated_balanced` 1,045. AUTHORED
values: resolved_by_dose (2,344) Â· singleton (2,260) Â· unvalidated_balanced (1,045)
example: CDH1 / hsa-miR-25-3p â†' unvalidated_balanced

---

arm_iqr
ARM · REAL Â‰¥ 0 · 69%

Interquartile range of the arm's abundance â€” its DYNAMIC RANGE. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 0.9

---

arm_lfc_HLY_NAT_raw
ARM · REAL, SIGNED · 63%

log-FC GTEx-healthy â†' TCGA-NAT, raw. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 9.49

---

arm_lfc_HLY_TUM_QN
ARM · REAL, SIGNED · 64%

log-FC GTEx-healthy â†' TCGA-tumour, on quantile-normalised values. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 4.79

---

arm_lfc_HLY_TUM_raw
ARM · REAL, SIGNED · 63%

As `arm_lfc_HLY_TUM_QN` but against the RAW GTEx median. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 9.46

---

arm_lfc_NAT_TUM
ARM · REAL, SIGNED · 68%

log-FC of the arm's median abundance, matched NAT â†' tumour. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' -0.03

---

arm_med_rpm
ARM · REAL Â‰¥ 0 · 69%

Median linear RPM of this ARM across the cohort. Arm-rung, so it REPEATS across the arm's genes. Fill 0.686 â€” the gap is arms with no expression record, not zeros. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 13.1

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

---

arm_pct_above_floor
ARM · PERCENT · 69%

Percent of samples in which the arm sits above the detection floor (median 99). Feeds `arm_spiker` together with `arm_iqr`. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 100

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

---

arm_role_in_family
ARM-IN-FAMILY · CATEGORICAL · 72%

â›” **The ARM's DOSE role inside its seed family â€” NOT the family's role in the gene**, which is what the name suggests. AUTHORED
values: sole (2,260) Â· co (1,045) Â· minor (395) Â· dominant (379)
example: CDH1 / hsa-miR-25-3p â†' co

---

arm_share_of_family_dose
ARM-IN-FAMILY · FRACTION 0Â€“1 · 69%

The arm's share of its seed family's TOTAL dose â€” denominator is the **full family**, not the design (median 0.69). The quantity `family_role` is cut from. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 0.56

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

arm_spiker

ARM BOOLEAN 69%

`arm_pct_floor < 40 AND arm_iqr > 1.5` â€” the arm sits at/below the detection floor in most samples yet swings widely when present. AUTHORED

values: False (3,706) Â· True (171)

example: CDH1 / hsa-miR-25-3p â†' False

BETA_ Â· 8

core â€" the columns findings rest on

8 columns Â· rung **edge** Â· coverage **20%â€"100%** (median 100%)
4× real â‰¥ 0 Â· 4× fraction 0â€"1

beta_arm

EDGE REAL â‰¥ 0 20%

The arm's OWN Î² from the within-cell fit (`arm_rung.py`), defined only in multi-arm cells (fill 0.203). AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.04066

beta_deconv

EDGE REAL â‰¥ 0 100%

Î² refit on the DECONV design (core C + the 8 non-malignant Wu-major lineages, Cancer-Epithelial held out). AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.1737

beta_frac

EDGE FRACTION 0â€"1 100%

MAGNITUDE share (beta_f / sum beta). AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.1575

beta_frac_abs

EDGE FRACTION 0â€"1 100%

|E[Î²_f]| / Î£|E[Î²]| â€" the share computed from the POINT ESTIMATES. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.1565

beta_frac_deconv

EDGE FRACTION 0â€"1 100%

`beta_frac` recomputed on the deconv design â€" the composition-adjusted share of the gene's budget. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.1411

beta_frac_sd

EDGE FRACTION 0â€"1 100%

Posterior SD of the per-draw fraction Î²_f/Î£Î². AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.03573

beta_sd

EDGE REAL â‰¥ 0 100%

Posterior summaries of the coupling coefficient from the dense Gibbs fit (`learned/readouts.py`). `_sd` is the posterior SD, `_frac` the share of the gene's total beta. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.04851

beta_sd_deconv

EDGE REAL â‰¥ 0 100%

Posterior SD on the deconv design. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.04548

N_ Â· 5

core â€" the columns findings rest on

5 columns Â· **spans edge, family, gene** â€" this prefix groups columns living on DIFFERENT units Â· coverage **20%â€"100%** (median 100%)
4× count Â· 1× constant

n_HLY_meas

EDGE COUNT 72%

How many of this gene's arms have a MEASURED healthy (GTEx) level â€" as opposed to one manufactured by the multi-mapping collapse. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 8

n_arm_in_cell

FAMILY | COUNT | 20%

How many arms sit in this edge's (gene, seed_family) CELL â€" i.e. how many of that family's members appear in THIS gene's design. AUTHORED

values: `2.0 (670) Â· 3.0 (258) Â· 4.0 (100) Â· 5.0 (55) Â· 6.0 (36) Â· 7.0 (28)`

example: `CDH1 / hsa-miR-25-3p â†' 2`

---

n_fam

GENE | COUNT | 100%

How many families compete at this edge's GENE. Gene-rung â€" it repeats across the gene's edges by construction. beta is mathematically inert where it is 1. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 25`

---

n_family_in_design

FAMILY | COUNT | 100%

Members of this arm's seed family **THAT ARE IN THIS GENE'S DESIGN** â€" 1 on **4,497 of 5,644 edges (79.7%)**. AUTHORED

values: `1.0 (4,497) Â· 2.0 (670) Â· 3.0 (258) Â· 4.0 (100) Â· 5.0 (55) Â· 6.0 (36) Â· 7.0 (28)`

example: `CDH1 / hsa-miR-25-3p â†' 2`

---

n_samples

GENE | CONSTANT | 100%

Samples the edge was fit on â€" CONSTANT (1,040) across the card. AUTHORED

values: `1040.0 (5,644)`

example: `CDH1 / hsa-miR-25-3p â†' 1040`

SHARE_ Â· 4

core â€" the columns findings rest on

4 columns Â· rung **edge** Â· coverage **46%â€"72%** (median 72%)
4× fraction 0â€"1

share_HLY

EDGE | FRACTION 0â€"1 | 72%

The arm's share of the gene's healthy (GTEx) pressure budget, median **0.129**. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.15`

---

share_HLY_meas

EDGE | FRACTION 0â€"1 | 46%

`share_HLY` over ONLY the arms GTEx genuinely measures â€" invisible arms ABSTAIN (NaN) instead of being zeroed (MH-210). Median **0.244** against the imputed 0.129. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.297`

---

share_NAT

EDGE | FRACTION 0â€"1 | 72%

The arm's share of its gene's pressure budget in matched normal (NAT). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.219`

---

share_TUM

EDGE | FRACTION 0â€"1 | 72%

The arm's share of its gene's pressure budget in tumour. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.243`

CTX_ Â· 18

core â€" the columns findings rest on

18 columns Â· **spans arm, gene** â€" this prefix groups columns living on DIFFERENT units Â· coverage **84%â€"100%** (median 99%)
4× real â‰¥ 0 Â· 4× signed âˆ'1â€¦1 Â· 3× real, signed Â· 3× count Â· 2× boolean Â· 1× categorical Â· 1× fraction 0â€"1

ctx_apriori_class

GENE | CATEGORICAL | 100%

The a-priori measurability class assigned to the gene's design before any fit (`gene_atlas` via `card_context`). Range **4 levels**, 100% filled. AUTHORED

values: `A_COMPETENT (3,989) Â· D_UNDEFINED (654) Â· B_PARTIAL (609) Â· C_WEAK (397)`

example: `CDH1 / hsa-miR-25-3p â†' A_COMPETENT`

---

ctx_arm_abundant
ARM | BOOLEAN | 95%

`ctx_arm_dose >= log2(11)` â€” the arm-abundance gate, ARM-rung on an edge card. See the arm card's entry. AUTHORED

values: `True (3,384) Â· False (1,981)`

example: `CDH1 / hsa-miR-25-3p â†' True`

> *Defined on: arms present on the EDGE card (729 of 2,450) â€” lifted arm-rung columns, names preserved*

---

ctx_arm_dose
ARM | REAL â‰¥ 0 | 95%

This arm's median log2(RPM+1) â€” see the arm card's entry. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 13.05`

---

ctx_ceiling
GENE | SIGNED âˆ'1â€¦1 | 99%

The UPPER BOUND on what ANY estimator can achieve for this gene â€” the 5-fold out-of-fold RÂ² of an UNRESTRICTED OLS on the gene's real design (`card_context.py:15`). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.1535`

---

ctx_collapse
GENE | REAL â‰¥ 0 | 100%

`n_arm / n_fam` for the gene â€” **how much curation BUNDLES same-seed mates onto it** (1â€“5). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 1.091`

---

ctx_d_collin
GENE | SIGNED âˆ'1â€¦1 | 84%

Within-design COLLINEARITY asymmetry, real minus decoy. The second design-match axis MH-167 tested; also bounded and not moving the gap. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' -0.02497`

> *Defined on: genes with >=2 seed families (a collinearity measure needs 2 columns)*

---

ctx_d_fam
GENE | REAL, SIGNED | 99%

Design-WIDTH asymmetry between the real design and its matched decoy (real minus fake families). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0`

---

ctx_dose_max
GENE | REAL â‰¥ 0 | 100%

The LARGEST regulator dose in the gene's design (`gene_atlas` via `card_context`). Range **0.215 â€“ 17.8**, 100% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 15.38`

---

ctx_dose_med
GENE | REAL â‰¥ 0 | 100%

The MEDIAN regulator dose in the gene's design (`gene_atlas` via `card_context`). Range **0.215 â€“ 17.8**, 100% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 6.318`

---

ctx_frac_abund
GENE | FRACTION 0â€“1 | 100%

FRACTION of the gene's regulators clearing the abundance gate (`gene_atlas` via `card_context`). Range **0 â€“ 1**, 100% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.5833`

---

ctx_gap_core
GENE | SIGNED âˆ'1â€¦1 | 99%

The Î²-vs-decoy MARGIN under the CORE confounder block (`gene_atlas` via `card_context`). Range **-0.476 â€“ +0.541**, 99% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' -0.2714`

---

ctx_gap_d_dose
GENE | REAL, SIGNED | 99%

Dose asymmetry between the real and decoy designs (median âˆ'0.43). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.1238`

---

ctx_gap_deconv
GENE | SIGNED âˆ'1â€¦1 | 99%

The same margin under the DECONVOLUTION block (`gene_atlas` via `card_context`). Range **-0.33 â€“ +0.374**, 99% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' -0.1864`

---

ctx_gap_retention
GENE | REAL, SIGNED | 95%

`gap_deconv / gap_core` â€” how much of the decoy margin survives adjustment (`gene_atlas` via `card_context`). Range **-16.5 â€“ 17**, 95% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.6867`

---

ctx_measurable
GENE | BOOLEAN | 99%

Is the gene's `ctx_ceiling` above `CEILING_ROBUST` (**0.02**, not 0 â€” MH-144)? AUTHORED

values: `True (4,189) Â· False (1,428)`

example: `CDH1 / hsa-miR-25-3p â†' True`

ctx_n_abund

GENE COUNT 100%

How many of the gene's regulators clear the abundance gate (`gene_atlas` via `card_context`). Range **0 â€“ 53**, 100% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†’ 7`

---

ctx_n_fam_abund

GENE COUNT 100%

Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†’ 6`

---

ctx_n_fam_multi

GENE COUNT 100%

How many of the gene's families hold MORE THAN ONE arm (`gene_atlas` via `card_context`). Range **0 â€“ 14**, 100% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†’ 1`

CAL_ Â· 7

core â€” the columns findings rest on

---

7 columns Â· rung **edge** Â· coverage **100%â€“100%** (median 100%)
4× real â‰¥ 0 Â· 3× boolean

cal_beta_sd

EDGE REAL â‰¥ 0 100%

The calibrated posterior SD of Î² (central estimate). Range **0.00101 â€“ 3.94**, 100% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†’ 0.05952`

---

cal_beta_sd_hi

EDGE REAL â‰¥ 0 100%

Upper end of the calibrated SD band. Range **0.00112 â€“ 4.39**, 100% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†’ 0.06636`

---

cal_beta_sd_lo

EDGE REAL â‰¥ 0 100%

Lower end of the calibrated SD band. Range **0.000912 â€“ 3.57**, 100% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†’ 0.05396`

---

cal_identified

EDGE BOOLEAN 100%

Is the edge identified (`|cal_z| > 2`) under the CALIBRATED posterior width â€” the honest version of `identified`: **1,110 vs 1,375 edges**, i.e. **19.9% [18.1â€“22.4]** against 24.8%. AUTHORED

values: `False (4,534) Â· True (1,110)`

example: `CDH1 / hsa-miR-25-3p â†’ True`

---

cal_identified_hi

EDGE BOOLEAN 100%

The UPPER end of the calibrated identification band â€” `|cal_z| > 2` under the optimistic calibration (990 edges vs 1,110 central). AUTHORED

values: `False (4,654) Â· True (990)`

example: `CDH1 / hsa-miR-25-3p â†’ True`

---

cal_identified_lo

EDGE BOOLEAN 100%

The LOWER end of the calibrated identification band (1,234 edges vs 1,110 central). See `cal_identified_hi` â€” the pair is the uncertainty, and reporting one alone hides it. AUTHORED

values: `False (4,410) Â· True (1,234)`

example: `CDH1 / hsa-miR-25-3p â†’ True`

---

cal_z

EDGE REAL â‰¥ 0 100%

`beta / cal_beta_sd` â€” the CALIBRATED z. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†’ 3.702`

ECHIM_ Â· 3

core â€” the columns findings rest on

---

3 columns Â· rung **edge** Â· coverage **8%â€“26%** (median 22%)
2× count Â· 1× real â‰¥ 0

echim_manakov_w
EDGE | REAL â‰¥ 0 | 22%

Manakov chimeric-eCLIP support weight for this edge (median 3, max 1,712). AUTHORED
example: ESR1 / hsa-miR-20b-5p â†' 0.3333

---

echim_n_sources
EDGE | COUNT | 26%

How many chimeric-eCLIP sources support this edge (1 on 1,151 Â· 2 on 262 Â· 3 on 34). AUTHORED
values: 1.0 (1,151) Â· 2.0 (262) Â· 3.0 (34) Â· 4.0 (2)
example: CDH1 / hsa-miR-92a-3p â†' 1

---

echim_tarbase_w
EDGE | COUNT | 8%

TarBase chimeric support weight (median 2, max 28) â€" a much shorter tail than the Manakov twin, and defined on only 8.4% of edges. AUTHORED
example: CDH1 / hsa-miR-92a-3p â†' 4

> *Defined on: edges with a chimeric (AGO-ligation) duplex â€" âš per-source, NEVER pooled; cross-tissue, not breast*

RANK_ Â· 4

core â€" the columns findings rest on

4 columns Â· rung **edge** Â· coverage **46%â€"72%** (median 72%)
4× count

rank_HLY
EDGE | COUNT | 72%

The arm's RANK within its gene's pressure budget in the GTEx-healthy state (1 = largest). AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 4

---

rank_HLY_meas
EDGE | COUNT | 46%

The arm's WITHIN-GENE rank by dose in one state. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 2

---

rank_NAT
EDGE | COUNT | 72%

The arm's RANK within its gene's pressure budget in matched normal (NAT) (1 = largest). AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 2

---

rank_TUM
EDGE | COUNT | 72%

The arm's RANK within its gene's pressure budget in tumour (1 = largest). AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 2

SHIFT_ Â· 1

core â€" the columns findings rest on

1 columns Â· rung **edge** Â· coverage **72%â€"72%** (median 72%)
1× categorical

shift_class
EDGE | CATEGORICAL | 72%

Cross-state class for the edge. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' tumour_realized

HEALTHY_ Â· 3

supporting â€" read these when the core raises a question

3 columns Â· rung **arm** Â· identical column set to the arm block Â· coverage **64%â€"72%** (median 64%)
1× categorical Â· 1× real â‰¥ 0 Â· 1× boolean

healthy_leg

ARM | CATEGORICAL | 72%

Provenance of this arm's GTEx healthy leg â€" `measured`, `collapse_blind` or `true_absent`. AUTHORED

values: `measured (2,573) Â· collapse_blind (854) Â· true_absent (652)`

example: `CDH1 / hsa-miR-25-3p â†' measured`

---

healthy_potential

ARM | REAL â‰¥ 0 | 64%

The arm's IMPUTED healthy abundance â€" a repression-POTENTIAL proxy, collapse-repaired (0.27 â€" 17.9). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 8.33`

---

healthy_uninformative

ARM | BOOLEAN | 64%

Is the healthy leg both `collapse_blind` AND high-potential â€" i.e. the one combination where an *acquired* call is genuinely unsafe? **True on 143 of 3,490 edges (4%).** AUTHORED

values: `False (3,490) Â· True (143)`

example: `CDH1 / hsa-miR-25-3p â†' False`

DOSE_ Â· 7

supporting â€" read these when the core raises a question

> **7** columns Â· **spans arm, edge** â€" this prefix groups columns living on DIFFERENT units Â· coverage **55%â€"72%** (median 72%)
> 3× count Â· 2× real â‰¥ 0 Â· 1× categorical Â· 1× constant

dose_class

EDGE | CATEGORICAL | 72%

Direction of the arm's NATâ†'tumour dose change â€" `gain` (1,620) / `flat` (1,286) / `loss` (1,173). Range **3 levels**, 72% filled. AUTHORED

values: `gain (1,620) Â· flat (1,286) Â· loss (1,173)`

example: `CDH1 / hsa-miR-25-3p â†' flat`

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

---

dose_comp_retention

ARM | REAL â‰¥ 0 | 55%

The arm's dose signal after conditioning on COMPOSITION, over its raw dose. â‰ˆ1.00 for real doses. Range **0.51 â€" 1.66**, 55% filled. AUTHORED

example: `CDH1 / hsa-miR-30a-5p â†' 1.02`

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

---

dose_confounded

ARM | CONSTANT | 72%

Is this edge's NATâ†'tumour dose a composition/proliferation artifact (either gated retention < 0.4)? AUTHORED

values: `False (4,079)`

example: `CDH1 / hsa-miR-25-3p â†' False`

---

dose_prolif_retention

ARM | REAL â‰¥ 0 | 55%

The same against PROLIFERATION. â‰ˆ1.00 for real doses â‡' the NATâ†'tumour acquisitions are cell-intrinsic. Range **0.69 â€" 1.25**, 55% filled. AUTHORED

example: `CDH1 / hsa-miR-30a-5p â†' 1.02`

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

---

dose_rank_HLY

EDGE | COUNT | 72%

The arm's within-gene DOSE rank (abundance only, no Î²) in healthy. Range **1 â€" 53**, 72% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 3`

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

---

dose_rank_NAT

EDGE | COUNT | 72%

Its within-gene dose rank in NAT. Range **1 â€" 57**, 72% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 2`

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

---

dose_rank_TUM

EDGE | COUNT | 72%

Its within-gene dose rank in tumour. Range **1 â€" 58**, 72% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 2`

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

IDENTITY_ Â· 5

supporting â€” read these when the core raises a question

5 columns Â· rung **edge** Â· coverage **96%â€“98%** (median 96%)
2× fraction 0â€“1 Â· 2× real, signed Â· 1× boolean

identity_abs
EDGE | FRACTION 0â€“1 | 96%

|identity| as a fraction of the gene's total |identity|, **clipped to [0,1]**. AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 0.3305

identity_allocated
EDGE | REAL, SIGNED | 96%

Family identity with phantom minor arms forced to 0 â€” the allocation actually used downstream. AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 0.3367

identity_coherence
EDGE | FRACTION 0â€“1 | 96%

`1 / Î£|identity|` over the gene, clipped to (0,1] â€” how much the gene's identity shares CANCEL. AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 0.9817

identity_deconv
EDGE | REAL, SIGNED | 98%

Shapley identity recomputed under the composition-adjusted design. AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 0.2929

identity_reliable
EDGE | BOOLEAN | 96%

Is this edge's Shapley `identity` trustworthy â€” `identity_coherence` â‰¥ 0.5 AND |identity| â‰¤ 1? AUTHORED

values: True (5,272) Â· False (159)

example: CDH1 / hsa-miR-25-3p â†’ True

D_RANK_ Â· 4

supporting â€” read these when the core raises a question

4 columns Â· rung **edge** Â· coverage **46%â€“72%** (median 72%)
4× real, signed

d_rank_HLY_NAT
EDGE | REAL, SIGNED | 72%

Change in the arm's WITHIN-GENE budget rank from the GTEx-healthy state to matched normal (NAT). AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 2

d_rank_HLY_TUM
EDGE | REAL, SIGNED | 72%

Change in the arm's WITHIN-GENE budget rank from the GTEx-healthy state to tumour. AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 2

d_rank_HLY_TUM_meas
EDGE | REAL, SIGNED | 46%

`d_rank_HLY_TUM` recomputed over arms with a MEASURED healthy level only â€” the collapse-safe version. Prefer it over the bare column. AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 0

> *Defined on: edges whose gene has â‰¥1 GTEx-MEASURED regulator â€” âš NaN = GTEx cannot see it (abstention), NOT 0*

d_rank_NAT_TUM
EDGE | REAL, SIGNED | 72%

Change in the arm's WITHIN-GENE budget rank from matched normal (NAT) to tumour. AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 0

GENE_ Â· 6

supporting â€” read these when the core raises a question

6 columns Â· rung **gene** Â· coverage **72%â€“100%** (median 99%)
2× categorical Â· 2× boolean Â· 1× real â‰¥ 0 Â· 1× real, signed

gene_cancer_role
GENE · CATEGORICAL · 100%

â›” **NOT the arm's role â€” the GENE's cancer role** (`oncogene` / `tsg` / `oncogene/tsg` / `unknown`), from `gene_roles.load_gene_roles` . AUTHORED

values: unknown (3,667) Â· oncogene (1,189) Â· tsg (661) Â· oncogene/tsg (132)

example: CDH1 / hsa-miR-25-3p â†' tsg

gene_dominated
GENE · BOOLEAN · 97%

Does one family carry **more than 60%** of this gene's tumour pressure budget (2,641 of 5,495)? AUTHORED

values: False (2,854) Â· True (2,641)

example: CDH1 / hsa-miR-25-3p â†' False

gene_iqr
GENE · REAL â‰¥ 0 · 72%

Interquartile range of the TARGET GENE's expression â€” its dynamic range. Range **0 â€“ 6.27**, 72% filled. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 1.72

> *Defined on: genes with a NAT-TUM gene-level leg*

gene_lfc_NAT_TUM
GENE · REAL, SIGNED · 72%

The target gene's own NATâ†'tumour log fold-change. Range **-5.85 â€“ 5.38**, 72% filled. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.73

> *Defined on: genes with a NAT-TUM gene-level leg*

gene_net_repr
GENE · BOOLEAN · 100%

Is this gene NET-REPRESSED in tumour by the Â§6b-RETIRED heuristic lane (1,376 of 5,634 edges)? AUTHORED

values: False (4,258) Â· True (1,376)

example: CDH1 / hsa-miR-25-3p â†' False

gene_repr_class
GENE · CATEGORICAL · 100%

The gene's repression class from the retired heuristic lane. Range **5 levels**, 100% filled. AUTHORED

values: never_repressed (2,939) Â· lost_repression (1,318) Â· constitutive_repressed (770) Â· gained_repression (606) Â· na (1)

example: CDH1 / hsa-miR-25-3p â†' never_repressed

AGO_ Â· 2

supporting â€” read these when the core raises a question

2 columns Â· rung **arm** Â· âš ago_loading is rank-identical to the arm card's ago_dom â€" one quantity, two names Â· coverage **95%â€“100%** (median 97%)
1× fraction 0â€“1 Â· 1× boolean

ago_loading
ARM · FRACTION 0â€“1 · 95%

AGO/RISC loading discount. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.996

ago_loading_measured
ARM · BOOLEAN · 100%

Was `ago_loading` actually MEASURED for this edge (vs defaulted to 1.0). True for 3,820/5,648 edges (67.6%). AUTHORED

values: True (3,820) Â· False (1,829)

example: CDH1 / hsa-miR-25-3p â†' True

> *Defined on: arms with a Manakov AGO-dominance measurement â€" â›" NOT loading(), which fabricates 1.0*

PIP_ Â· 2

reference â€” context, provenance and rarely-quoted detail

2 columns Â· rung **edge** Â· coverage **100%â€“100%** (median 100%)
1× constant Â· 1× fraction 0â€“1

pip_dense
EDGE | CONSTANT | 100%

âš CONSTANT by construction â€” the coupling readout is called with Ï€â‰¡1, so every edge has the same value. AUTHORED

values: `1.0 (5,644)`

example: `CDH1 / hsa-miR-25-3p â†' 1`

pip_discovery
EDGE | FRACTION 0â€“1 | 100%

Posterior inclusion probability. `pip_dense` comes from the pi=1 coupling readout; `pip_discovery` from the evidence-pi readout. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 1`

RETENTION_ Â· 3

reference â€” context, provenance and rarely-quoted detail

3 columns Â· rung **edge** Â· coverage **72%â€“100%** (median 100%)
1× real â‰¥ 0 Â· 1× boolean Â· 1× real, signed

retention_beta
EDGE | REAL â‰¥ 0 | 100%

Fraction of an effect surviving covariate adjustment (adjusted / raw). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.7885`

retention_reliable
EDGE | BOOLEAN | 100%

Is this edge's `retention` ratio trustworthy â€” i.e. does it have a real denominator? AUTHORED

values: `False (4,269) Â· True (1,375)`

example: `CDH1 / hsa-miR-25-3p â†' True`

retention_rho
EDGE | REAL, SIGNED | 72%

Ï_deconv / Ï_core â€” AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.65`

OOF_ Â· 3

reference â€” context, provenance and rarely-quoted detail

3 columns Â· rung **family** Â· coverage **20%â€“20%** (median 20%)
3× signed âˆ'1â€¦1

oof_drho
FAMILY | SIGNED âˆ'1â€¦1 | 20%

Arm-level OOF rho MINUS family-level. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' -0.004735`

oof_rho_arm
FAMILY | SIGNED âˆ'1â€¦1 | 20%

Out-of-fold coupling from the ARM-grain fit. Range **-0.542 â€“ +0.324**, 20% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' -0.2411`

> *Defined on: multi-arm cells only (n_arm_in_cell > 1) â€” 20.3% of edges*

oof_rho_fam
FAMILY | SIGNED âˆ'1â€¦1 | 20%

Out-of-fold coupling from the FAMILY-grain fit â€” their difference is `oof_drho`, the arm-resolution gain. Range **-0.545 â€“ +0.309**, 20% filled. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' -0.2364`

> *Defined on: multi-arm cells only (n_arm_in_cell > 1) â€” 20.3% of edges*

GRANK_ Â· 3

reference â€” context, provenance and rarely-quoted detail

3 columns Â· rung **arm** Â· coverage **63%â€“69%** (median 68%)
3× count

grank_HLY
ARM COUNT 63%

The arm's GLOBAL abundance percentile across the whole cohort in the GTEx-healthy state. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 96

grank_NAT
ARM COUNT 68%

The arm's GLOBAL abundance percentile across the whole cohort in matched normal (NAT). AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 99

grank_TUM
ARM COUNT 69%

The arm's GLOBAL abundance percentile across the whole cohort in tumour. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 99

DGLOBAL_ Â· 3

reference â€” context, provenance and rarely-quoted detail

3 columns Â· rung **arm** Â· coverage **63%â€“68%** (median 63%)
3× real, signed

dGlobal_HLY_NAT
ARM REAL, SIGNED 63%

Change in the arm's GLOBAL (cohort-wide) abundance percentile from the GTEx-healthy state to matched normal (NAT). AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 3

dGlobal_HLY_TUM
ARM REAL, SIGNED 63%

Change in the arm's GLOBAL (cohort-wide) abundance percentile from the GTEx-healthy state to tumour. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 3

dGlobal_NAT_TUM
ARM REAL, SIGNED 68%

Change in the arm's GLOBAL (cohort-wide) abundance percentile from matched normal (NAT) to tumour. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 0

OWN_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **arm** Â· coverage **88%â€“88%** (median 88%)
1× real â‰¥ 0

own_specific_frac
ARM REAL Â‰¥ 0 88%

Fraction of the arm's variance that is target-specific. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 0.55

COMPOSITION_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **100%â€“100%** (median 100%)
1× categorical

composition_class
EDGE CATEGORICAL 100%

Composition-adjustment verdict for this row (e.g. `composition_explained` when the effect does not survive the deconvolution block). AUTHORED

values: `cell_intrinsic (3,764) Â· partial (1,167) Â· composition_explained (713)`

example: `CDH1 / hsa-miR-25-3p â†' cell_intrinsic`

W_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **100%â€"100%** (median 100%)
1× real â‰¥ 0

w_max
GENE REAL â‰¥ 0 100%

Maximum curated EVIDENCE weight over the gene's regulators. Gene-rung repeat. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 13.24`

M_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **100%â€"100%** (median 100%)
1× real â‰¥ 0

m_nnls
EDGE REAL â‰¥ 0 100%

Bagged-NNLS weight. Exactly 0 in 31.7% of families â€” this is what lets `identity` say a regulator contributes nothing, which `beta` structurally cannot. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.2293`

TERM_ Â· 3

reference â€” context, provenance and rarely-quoted detail

3 columns Â· rung **edge** Â· coverage **68%â€"68%** (median 68%)
2× signed âˆ'1â€¦1 Â· 1× real, signed

term_ABUND
EDGE SIGNED âˆ'1â€¦1 68%

Decomposition of the acquisition score into ABUNDANCE, WIRING and INTERACTION terms. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' -0.004`

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

---

term_INTERACT
EDGE SIGNED âˆ'1â€¦1 68%

The INTERACTION term â€” the part of the NATâ†'tumour change attributable to abundance and wiring moving TOGETHER. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' -0.003`

---

term_WIRING
EDGE REAL, SIGNED 68%

The WIRING term of the dose-vs-wiring decomposition: how much of the NATâ†'tumour change comes from the arm's M-weight moving rather than its abundance. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 1.376`

PRIOR_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **100%â€"100%** (median 100%)
1× constant

prior_pi
EDGE CONSTANT 100%

The Gibbs sampler's inclusion prior â€” **CONSTANT 0.05** across the card. AUTHORED
values: 0.05 (5,644)
example: CDH1 / hsa-miR-25-3p â†' 0.05

EDGE_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **73%â€"73%** (median 73%)
1× signed âˆ'1â€¦1

edge_rho_adj
EDGE SIGNED âˆ'1â€¦1 73%

An edge-rung quantity carried onto this row. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' -0.198

> *Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)*

ESUB_ Â· 8

reference â€" context, provenance and rarely-quoted detail

8 columns Â· rung **edge** Â· coverage **98%â€"100%** (median 99%)
6× signed âˆ'1â€¦1 Â· 1× categorical Â· 1× p-value

esub_best_sub
EDGE CATEGORICAL 100%

WHICH subtype gave the strongest coupling. AUTHORED
values: Her2 (1,782) Â· Basal (1,679) Â· LumB (1,178) Â· LumA (1,005)
example: CDH1 / hsa-miR-25-3p â†' Her2

esub_q_adj
EDGE P-VALUE 98%

BH-adjusted q for the best-subtype contrast â€" AUTHORED
example: CDH1 / hsa-miR-25-3p â†' 0.3849

esub_rho_Basal
EDGE SIGNED âˆ'1â€¦1 99%

This edge's coupling computed WITHIN the Basal PAM50 subtype. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' -0.1683

esub_rho_Her2
EDGE SIGNED âˆ'1â€¦1 98%

This edge's coupling computed WITHIN the Her2 PAM50 subtype. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' -0.3531

esub_rho_LumA
EDGE SIGNED âˆ'1â€¦1 99%

This edge's coupling computed WITHIN the LumA PAM50 subtype. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' -0.174

esub_rho_LumB
EDGE SIGNED âˆ'1â€¦1 99%

This edge's coupling computed WITHIN the LumB PAM50 subtype. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' -0.1127

esub_rho_best
EDGE SIGNED âˆ'1â€¦1 100%

Coupling in the arm's BEST PAM50 subtype. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' -0.3531

esub_rho_pool
EDGE SIGNED âˆ'1â€¦1 100%

Coupling pooled across subtypes â€" the unselected reference for `esub_rho_best` (median âˆ'0.0135, indistinguishable from a random-subtype draw at âˆ'0.0123). AUTHORED
example: CDH1 / hsa-miR-25-3p â†' -0.1909

MEAN_ Â· 2

reference â€” context, provenance and rarely-quoted detail

2 columns Â· rung **arm** Â· coverage **90%â€“90%** (median 90%)
2× real, signed

mean_dGlobalRank
ARM · REAL, SIGNED · 90%

Mean change in the arm's GLOBAL abundance percentile across the paired patients. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.01

mean_own_shift
ARM · REAL, SIGNED · 90%

A per-row mean of a lower-rung quantity; check `agg_of` for which rung it summarises. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' -0.194

> *Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)*

SURROGATE_ Â· 2

reference â€” context, provenance and rarely-quoted detail

2 columns Â· rung **arm** Â· coverage **3%â€“3%** (median 3%)
1× fraction 0â€“1 Â· 1× categorical

surrogate_corr
ARM · FRACTION 0â€“1 · 3%

How well the surrogate tracks the arm where both are observed (0.54 â€“ 0.89). AUTHORED

values: 0.874 (43) Â· 0.892 (35) Â· 0.621 (29) Â· 0.842 (26) Â· 0.541 (23) Â· 0.606 (18) Â· 0.728 (16)

example: CDH1 / hsa-miR-30a-5p â†' 0.606

surrogate_instrument
ARM · CATEGORICAL · 3%

Which SURROGATE stands in for a collapsed GTEx healthy leg (7 levels; defined on 3% of edges). AUTHORED

values: hsa-miR-20a-5p (43) Â· hsa-miR-181b-5p (35) Â· hsa-miR-93-5p (29) Â· hsa-miR-429 (26) Â· hsa-let-7b-5p (23) Â· hsa-miR-30c-5p (18) Â· hsa-miR-221-3p (16)

example: CDH1 / hsa-miR-30a-5p â†' hsa-miR-30c-5p

CELL_ Â· 3

reference â€” context, provenance and rarely-quoted detail

3 columns Â· rung **family** Â· coverage **20%â€“20%** (median 20%)
2× real â‰¥ 0 Â· 1× boolean

cell_arm_sep_z
FAMILY · REAL â‰¥ 0 · 20%

`cell_beta_spread` in units of the cell's pooled posterior SD â€” **can these same-seed arms be told apart at all?** AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 3.196

cell_arms_resolvable
FAMILY · BOOLEAN · 20%

Are the cell's arms separable AND does splitting them help out-of-fold. True for 31.7% of edges in multi-arm cells. AUTHORED

values: False (783) Â· True (364)

example: CDH1 / hsa-miR-25-3p â†' True

cell_beta_spread
FAMILY · REAL â‰¥ 0 · 20%

max âˆ' min Î² across the arms in this family CELL â€” how far apart the cell's members are estimated to be. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.03

CPTAC_ Â· 20

reference â€" context, provenance and rarely-quoted detail

20 columns Â· rung **edge** Â· per-edge; the gene card aggregates it Â· coverage **45%â€"79%** (median 65%)
12× signed âˆ'1â€¦1 Â· 4× real, signed Â· 2× constant Â· 2× categorical

cptac_prosp_comp_class_prot
EDGE CATEGORICAL 59%

Composition class of this edge's PROTEIN coupling (cell_intrinsic / partial / composition_explained) in the prospective cohort. AUTHORED

values: composition_explained (1,427) Â· cell_intrinsic (1,278) Â· partial (647)

example: CDH1 / hsa-miR-25-3p â†' composition_explained

---

cptac_prosp_n
EDGE CONSTANT 79%

Patients in the prospective CPTAC cohort â€" **CONSTANT 101**. AUTHORED

values: 101.0 (4,486)

example: CDH1 / hsa-miR-25-3p â†' 101

---

cptac_prosp_ret_prot
EDGE REAL, SIGNED 59%

RETENTION at EDGE rung â€" this edge's composition-ADJUSTED protein coupling divided by its UNADJUSTED one, in the CPTAC prospective cohort. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.2189

---

cptac_prosp_ret_rna
EDGE REAL, SIGNED 61%

As `cptac_prosp_ret_prot` but against **mRNA** rather than protein: adjusted Ã· unadjusted mRNA coupling for this edge, prospective cohort. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.3

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_prosp_rho_disc
EDGE SIGNED âˆ'1â€¦1 78%

Spearman coupling of this single edge's arm against the discovery/decoy layer in the PROSPECTIVE cohort (independent patients, TMT-11, nâ‰ˆ101), adjusted for the C block (composition + target CN + proliferation). AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.08776

---

cptac_prosp_rho_disc_raw
EDGE SIGNED âˆ'1â€¦1 78%

Spearman coupling of this single edge's arm against the discovery/decoy layer in the PROSPECTIVE cohort (independent patients, TMT-11, nâ‰ˆ101), **UNADJUSTED** â€" no confounder block. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.014

---

cptac_prosp_rho_prot
EDGE SIGNED âˆ'1â€¦1 78%

Spearman coupling of this single edge's arm against PROTEIN in the PROSPECTIVE cohort (independent patients, TMT-11, nâ‰ˆ101), adjusted for the C block (composition + target CN + proliferation). AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.03748

---

cptac_prosp_rho_prot_raw
EDGE SIGNED âˆ'1â€¦1 79%

UNADJUSTED protein coupling, prospective. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.1712

---

cptac_prosp_rho_rna
EDGE SIGNED âˆ'1â€¦1 79%

Spearman coupling of this single edge's arm against mRNA in the PROSPECTIVE cohort (independent patients, TMT-11, nâ‰ˆ101), adjusted for the C block (composition + target CN + proliferation). AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.06188

---

cptac_prosp_rho_rna_raw
EDGE SIGNED âˆ'1â€¦1 79%

Spearman coupling of this single edge's arm against mRNA in the PROSPECTIVE cohort (independent patients, TMT-11, nâ‰ˆ101), **UNADJUSTED** â€" no confounder block. Range **-0.578 â€" +0.692**, 79% filled. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.2063

---

cptac_t105_comp_class_prot
EDGE CATEGORICAL 45%

Composition class of this row's PROTEIN coupling in the TCGA-OVERLAP cohort (n=105) â€" `cell_intrinsic`, `partial` or `composition_explained`. AUTHORED

values: cell_intrinsic (1,447) Â· composition_explained (632) Â· partial (463)

example: CDH1 / hsa-miR-25-3p â†' composition_explained

cptac_t105_n

EDGE | CONSTANT | 70%

Patients in the TCGA-overlap CPTAC cohort â€” **CONSTANT 105**. See the prospective twin. AUTHORED

values: 105.0 (3,968)

example: CDH1 / hsa-miR-25-3p â†' 105

---

cptac_t105_ret_prot

EDGE | REAL, SIGNED | 45%

Retention at edge rung on the TCGA-overlap cohort â€” see the prospective twin. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' -0.5722

---

cptac_t105_ret_rna

EDGE | REAL, SIGNED | 49%

As the prospective mRNA twin, on the TCGA-overlap cohort. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.6976

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_rho_disc

EDGE | SIGNED Â^'1Â€¦1 | 64%

Spearman coupling of this single edge's arm against the discovery/decoy layer in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.05624

---

cptac_t105_rho_disc_raw

EDGE | SIGNED Â^'1Â€¦1 | 65%

Spearman coupling of this single edge's arm against the discovery/decoy layer in the TCGA-OVERLAP cohort (n=105), **UNADJUSTED** â€” no confounder block. Range **-0.643 â€“ +0.727**, 65% filled. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' -0.03572

---

cptac_t105_rho_prot

EDGE | SIGNED Â^'1Â€¦1 | 64%

Spearman coupling of this single edge's arm against PROTEIN in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.0313

---

cptac_t105_rho_prot_raw

EDGE | SIGNED Â^'1Â€¦1 | 65%

Spearman coupling of this single edge's arm against PROTEIN in the TCGA-OVERLAP cohort (n=105), **UNADJUSTED** â€” no confounder block. Range **-0.714 â€“ +0.736**, 65% filled. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' -0.0547

---

cptac_t105_rho_rna

EDGE | SIGNED Â^'1Â€¦1 | 64%

Spearman coupling of this single edge's arm against mRNA in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). Range **-0.593 â€“ +0.684**, 64% filled. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' -0.06411

---

cptac_t105_rho_rna_raw

EDGE | SIGNED Â^'1Â€¦1 | 65%

Spearman coupling of this single edge's arm against mRNA in the TCGA-OVERLAP cohort (n=105), **UNADJUSTED** â€” no confounder block. Range **-0.782 â€“ +0.695**, 65% filled. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' -0.0919

REALIZATION_ Â· 2

reference â€” context, provenance and rarely-quoted detail

> **2** columns Â· rung **edge** Â· coverage **58%â€“69%** (median 63%)
> 1× fraction 0â€“1 Â· 1× real, signed

realization_identity

EDGE | FRACTION 0Â€“1 | 58%

Within-patient realization quantities â€” the score, its identity allocation, and who owns it. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.849

> *Defined on: genes with a family-level realization fit*

---

realization_score

EDGE | REAL, SIGNED | 69%

`â^'coupling_z_tum` â€” the calibrated tumour repression strength as a positive score, so bigger = more repressed. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 3.04

KD_ Â· 2

reference â€” context, provenance and rarely-quoted detail

2 columns Â· rung **edge** Â· coverage **85%â€“85%** (median 85%)
1× percent Â· 1× real, signed

kd_affinity_pct
EDGE PERCENT 85%

Is this gene among the arm's strongest predicted targets â€” a WITHIN-ARM percentile, **higher = stronger target**. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.8387

kd_repression
EDGE REAL, SIGNED 85%

scanMiR RBNS binding affinity. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' -0.8171

WIRING_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **68%â€“68%** (median 68%)
1× fraction 0â€“1

wiring_frac
EDGE FRACTION 0â€“1 68%

Fraction of the acquisition score attributable to WIRING rather than abundance. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 1

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

SD_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **20%â€“20%** (median 20%)
1× real â‰¥ 0

sd_arm
EDGE REAL â‰¥ 0 20%

Posterior SD of the arm-level Î². AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.01097

Z_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **20%â€“20%** (median 20%)
1× real â‰¥ 0

z_arm
EDGE REAL â‰¥ 0 20%

`beta_arm / sd_arm` from the WITHIN-CELL fit (`arm_rung.py`) â€” the arm's z inside its (gene, seed_family) cell only, defined on the 20.3% of edges in multi-arm cells. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 3.706

ACQUISITION_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **59%â€“59%** (median 59%)
1× real, signed

acquisition_score

EDGE | REAL, SIGNED | 59%

Composite acquisition score across states. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 3.35

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

DSHARE_ Â· 2

reference â€" context, provenance and rarely-quoted detail

---

2 columns Â· rung **edge** Â· coverage **65%â€"65%** (median 65%)
2× signed âˆ'1â€¦1

dShare_M_own

EDGE | SIGNED Âˆ'1Â€¦1 | 65%

Mean over the 103 paired patients of this arm's change in M-WEIGHTED share of its gene's total regulator dose, tumour minus that patient's OWN NAT. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.045

---

dShare_raw_own

EDGE | SIGNED Âˆ'1Â€¦1 | 65%

As `dShare_M_own` but on RAW abundance share, with the model's `M` weights removed. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.04

HEUR_ Â· 1

reference â€" context, provenance and rarely-quoted detail

---

1 columns Â· rung **gene** Â· coverage **100%â€"100%** (median 100%)
1× count

heur_gene_nreg

GENE | COUNT | 100%

Regulator count for this gene from the Â§6b-RETIRED heuristic lane. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 12

FAMILY_ Â· 1

reference â€" context, provenance and rarely-quoted detail

---

1 columns Â· rung **family** Â· coverage **69%â€"69%** (median 69%)
1× signed âˆ'1â€¦1

family_rho_adj

FAMILY | SIGNED Âˆ'1Â€¦1 | 69%

Family-level properties carried onto this row (size, the arm's dose share of it, and its role). AUTHORED

example: CDH1 / hsa-miR-25-3p â†' -0.332

HLY_ Â· 1

reference â€" context, provenance and rarely-quoted detail

---

1 columns Â· rung **edge** Â· coverage **61%â€"61%** (median 61%)
1× boolean

hly_leg_concordant

EDGE | BOOLEAN | 61%

Do the two HLYâ†'TUM legs (`_QN` and `_raw`) at least AGREE ON SIGN. **Concordant 76.1%, DISCORDANT 23.9%** of the 3,433 edges carrying both. AUTHORED

values: True (2,611) Â· False (822)

example: CDH1 / hsa-miR-25-3p â†' True

IS_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **58%â€“58%** (median 58%)
1× boolean

is_realization_owner

EDGE BOOLEAN 58%

A boolean flag; read the suffix. AUTHORED

values: False (2,498) Â· True (759)

example: CDH1 / hsa-miR-25-3p â†’ True

> *Defined on: genes with a family-level realization fit*

realization/gene_card.tsv

gene

One row per the gene's total incoming regulation. One row per gene, so nothing here is checkable by invariance â€” `agg_of` carries the meaning instead.

135 columns **1,549** rows **29** blocks median coverage **83%**

Contents â€” 29 blocks, most important first

N_ Â· 7

core â€” the columns findings rest on

7 columns Â· rung **gene** Â· coverage **91%â€“100%** (median 100%)
7× count

n_arms
GENE COUNT AGG ARM 100%

Arms in the LEARNED MODEL's design â€” `X.shape[1]` from `assemble_gene`, carried via `gene_atlas`. **This is the width the Gibbs actually fit** (median 2, max 90). AUTHORED

example: CDH1 â†' 26

n_cell_intrinsic
GENE COUNT 100%

FAMILIES whose coupling SURVIVES the composition block (retention â‰¥ 0.7). AUTHORED

example: CDH1 â†' 22

n_composition_explained
GENE COUNT 100%

FAMILIES whose coupling does NOT survive the composition block (retention < 0.4). AUTHORED

values: 0 (1,133) Â· 1 (286) Â· 2 (80) Â· 3 (26) Â· 4 (13) Â· 6 (7) Â· 5 (2) Â· 8 (2)

example: CDH1 â†' 0

n_fam
GENE COUNT AGG FAMILY 100%

How many seed families regulate this gene. AUTHORED

example: CDH1 â†' 25

n_hallmark_sets
GENE COUNT 91%

How many MSigDB Hallmark programs contain this gene (1â€“10, median 2). Membership only â€” it says nothing about regulation. AUTHORED

example: CDH1 â†' 3

n_identified
GENE | COUNT | 100%

Families reaching |z| > 2 on the UNCALIBRATED SD. **Zero for 691 genes (44.6%)** â€” meaning the model cannot distinguish ANY of that gene's families from zero. AUTHORED

example: `CDH1 â†’ 4`

---

n_pip_disc_gt50
GENE | COUNT | 100%

Count of the gene's families with `pip_discovery > 0.5` under the evidence-Ï€ readout. AUTHORED

example: `CDH1 â†’ 4`

(BARE) Â· 2

core â€” the columns findings rest on

---

2 columns Â· **spans gene, key** â€” this prefix groups columns living on DIFFERENT units Â· coverage **100%â€“100%** (median 100%)
1× fraction 0â€“1 Â· 1× identifier

concentration
GENE | FRACTION 0â€“1 | 100%

`beta.max() / beta.sum()` â€” the TOP FAMILY's share of the gene's total Î²; median **0.954**, i.e. most genes are dominated by one family. AUTHORED

example: `CDH1 â†’ 0.191`

---

gene
KEY | IDENTIFIER | 100%

KEY â€” HGNC gene symbol. AUTHORED

example: `JUN`

COMP_ Â· 16

core â€” the columns findings rest on

---

16 columns Â· rung **gene** Â· coverage **59%â€“90%** (median 73%)
6× categorical Â· 6× signed âˆ’1â€¦1 Â· 3× real, signed Â· 1× real â‰¥ 0

comp_cptac_prot_driver
GENE | CATEGORICAL | 59%

WHICH deconvolved compartment most explains this gene's PROTEIN coupling â€” `prolif` 219 Â· `CAFs` 173 Â· `purity` 140 Â· `PVL` 77 Â· `Endothelial` 67. AUTHORED

example: `CDH1 â†’ T-cells`

---

comp_cptac_prot_driver_ret
GENE | REAL, SIGNED | 59%

`1 âˆ' comp_cptac_prot_driver_share`, exactly. Median +0.398 raw, **+0.469** gated to |Ï_raw| â‰¥ 0.10 â€” quote the gated value. See the mRNA twin. AUTHORED

example: `CDH1 â†’ 0.5085`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_cptac_prot_driver_share
GENE | REAL, SIGNED | 59%

Protein-layer twin of `comp_tcga_mrna_driver_share`. Same exact complementarity with `_driver_ret` (8.9e-16 on 918 rows). AUTHORED

example: `CDH1 â†’ 0.4915`

---

comp_cptac_prot_rho_raw
GENE | SIGNED âˆ'1â€¦1 | 73%

Unadjusted aggregate PROTEIN coupling â€” denominator of `comp_cptac_prot_driver_ret`. AUTHORED

example: `CDH1 â†’ -0.1416`

---

comp_cptac_prot_top_budget
GENE | CATEGORICAL | 73%

The compartment the gene's miRNA BUDGET loads on most, on CPTAC protein. Range **10 levels**, 73% filled. AUTHORED

example: `CDH1 â†’ Normal Epithelial`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_cptac_prot_top_budget_r
GENE | SIGNED Âˆ'1Â€¦1 | 73%

As the TCGA mRNA twin, on CPTAC protein â€” the budget-vs-compartment correlation. Median **âˆ'0.118**, i.e. it runs the OPPOSITE way to mRNA (+0.121). AUTHORED

example: `CDH1 â†’ 0.3126`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

comp_cptac_prot_top_target
GENE | CATEGORICAL | 73%

The compartment the TARGET's expression loads on most. Range **10 levels**, 73% filled. AUTHORED

example: `CDH1 â†' T-cells`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_cptac_prot_top_target_r
GENE | SIGNED Âˆ'1Â€¦1 | 73%

As the TCGA mRNA twin, on CPTAC protein â€" the target-vs-compartment correlation. Median **+0.203**. AUTHORED

example: `CDH1 â†' -0.2385`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_tcga_mrna_driver
GENE | CATEGORICAL | 66%

WHICH compartment drives the gene's mRNA coupling â€" the one whose removal moves it most (`CAFs` 279 Â· `purity` 174 Â· `prolif` 148 Â· `Endothelial` 122 Â· `Myeloid` 109). AUTHORED

example: `CDH1 â†' T-cells`

---

comp_tcga_mrna_driver_ret
GENE | REAL, SIGNED | 66%

`1 âˆ' comp_tcga_mrna_driver_share` , exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment. AUTHORED

example: `CDH1 â†' 0.8681`

---

comp_tcga_mrna_driver_share
GENE | REAL Â‰¥ 0 | 66%

Share of the gene's mRNA coupling attributable to the single compartment that most changes it â€" **axiom 8's composition GATE**, the axis that separates the harm tails at p=0.00999 where the ensemble axis gives p=0.10. AUTHORED

example: `CDH1 â†' 0.1319`

---

comp_tcga_mrna_rho_raw
GENE | SIGNED Âˆ'1Â€¦1 | 90%

The gene's UNADJUSTED aggregate mRNA coupling â€" the DENOMINATOR of `comp_tcga_mrna_driver_ret` . AUTHORED

example: `CDH1 â†' -0.1921`

---

comp_tcga_mrna_top_budget
GENE | CATEGORICAL | 90%

The compartment the gene's miRNA BUDGET loads on most â€" a property of the REGULATORS. AUTHORED

example: `CDH1 â†' Normal Epithelial`

---

comp_tcga_mrna_top_budget_r
GENE | SIGNED Âˆ'1Â€¦1 | 90%

Correlation between the gene's **miRNA PRESSURE BUDGET** and the fraction of the compartment that budget loads on most (`comp_tcga_mrna_top_budget`), in TCGA mRNA. AUTHORED

example: `CDH1 â†' 0.2545`

---

comp_tcga_mrna_top_target
GENE | CATEGORICAL | 90%

The compartment the TARGET GENE's expression loads on most. AUTHORED

example: `CDH1 â†' prolif`

---

comp_tcga_mrna_top_target_r
GENE | SIGNED Âˆ'1Â€¦1 | 90%

Correlation between the **TARGET GENE's own expression** and the fraction of the compartment it loads on most. AUTHORED

example: `CDH1 â†' 0.3289`

CTX_ Â· 16

core â€" the columns findings rest on

16 columns Â· rung **gene** Â· SUBSET of the edge block, identical rungs Â· coverage **45%â€"100%** (median 90%)
4× signed â'1â€¦1 Â· 3× real â‰¥ 0 Â· 3× real, signed Â· 3× count Â· 1× categorical Â· 1× fraction 0â€"1 Â· 1× boolean

ctx_apriori_class
GENE | CATEGORICAL | 100%

A-priori competence class from the gene atlas (A_COMPETENT / B_PARTIAL / C_WEAK / D_UNDEFINED). AUTHORED

values: `D_UNDEFINED (730) Â· A_COMPETENT (428) Â· B_PARTIAL (295) Â· C_WEAK (96)`

example: `CDH1 â†' A_COMPETENT`

ctx_ceiling
GENE | SIGNED Â^'1Â€¦1 | 98%

The UPPER BOUND on what ANY estimator can achieve for this gene â€" the 5-fold out-of-fold RÂ² of an UNRESTRICTED OLS on the gene's real design (`card_context.py:15`). AUTHORED
example: `CDH1 â†' 0.1535`

---

ctx_collapse
GENE | REAL Â‰¥ 0 | 90%

`n_arm / n_fam` for the gene â€" **how much curation BUNDLES same-seed mates onto it** (1â€"5). AUTHORED
example: `CDH1 â†' 1.091`

---

ctx_d_collin
GENE | SIGNED Â^'1Â€¦1 | 45%

Mean pairwise COLLINEARITY among the gene's regulators â€" how much they move together (`gene_atlas` via `card_context`). Range **-0.699 â€" +0.495**, 45% filled. AUTHORED
example: `CDH1 â†' -0.02497`

---

ctx_d_fam
GENE | REAL, SIGNED | 87%

Design width at FAMILY grain (`gene_atlas` via `card_context`). Range **-2 â€" 25**, 87% filled. AUTHORED
example: `CDH1 â†' 0`

---

ctx_dose_max
GENE | REAL Â‰¥ 0 | 90%

The LARGEST regulator dose in the gene's design (`gene_atlas` via `card_context`). Range **0.215 â€" 17.8**, 90% filled. AUTHORED
example: `CDH1 â†' 15.38`

---

ctx_dose_med
GENE | REAL Â‰¥ 0 | 90%

The MEDIAN regulator dose in the gene's design (`gene_atlas` via `card_context`). Range **0.215 â€" 17.8**, 90% filled. AUTHORED
example: `CDH1 â†' 6.318`

---

ctx_frac_abund
GENE | FRACTION 0Â€"1 | 90%

FRACTION of the gene's regulators clearing the abundance gate (`gene_atlas` via `card_context`). Range **0 â€" 1**, 90% filled. AUTHORED
example: `CDH1 â†' 0.5833`

---

ctx_gap_core
GENE | SIGNED Â^'1Â€¦1 | 87%

Real-minus-decoy coupling gap for this gene. Pooled it is ~-0.012, but the per-gene IQR STRADDLES ZERO â€" this is a set-level distributional shift, not a per-gene effect. Do not rank genes by it. AUTHORED
example: `CDH1 â†' -0.2714`

---

ctx_gap_d_dose
GENE | REAL, SIGNED | 87%

Difference in dose between the real design and its decoy. Range **-5.41 â€" 1.74**, 87% filled. AUTHORED
example: `CDH1 â†' 0.1238`

---

ctx_gap_deconv
GENE | SIGNED Â^'1Â€¦1 | 87%

The same margin under the DECONVOLUTION block (`gene_atlas` via `card_context`). Range **-0.33 â€" +0.374**, 87% filled. AUTHORED
example: `CDH1 â†' -0.1864`

---

ctx_gap_retention
GENE | REAL, SIGNED | 83%

`gap_deconv / gap_core` â€" how much of the decoy margin survives adjustment (`gene_atlas` via `card_context`). Range **-16.5 â€" 17**, 83% filled. AUTHORED
example: `CDH1 â†' 0.6867`

---

ctx_measurable
GENE | BOOLEAN | 98%

Is the gene's `ctx_ceiling` above `CEILING_ROBUST` (**0.02**, not 0 â€" MH-144)? AUTHORED
values: `False (845) Â· True (672)`
example: `CDH1 â†' True`

---

ctx_n_abund
GENE | COUNT | 90%

How many of the gene's regulators clear the abundance gate (`gene_atlas` via `card_context`). Range **0 â€" 53**, 90% filled. AUTHORED
example: `CDH1 â†' 7`

---

ctx_n_fam_abund
GENE | COUNT | 90%

How many of the gene's families clear the abundance gate. Range **0 â€" 34**, 90% filled. AUTHORED
example: `CDH1 â†' 6`

---

ctx_n_fam_multi

GENE | COUNT | 90%

How many of the gene's families hold MORE THAN ONE arm (`gene_atlas` via `card_context`). Range **0 â€“ 14**, 90% filled. AUTHORED

example: `CDH1 â†' 1`

TOP_ Â· 8

supporting â€" read these when the core raises a question

8 columns Â· rung **gene** Â· coverage **78%â€"100%** (median 89%)
3× identifier Â· 2× real â‰¥ 0 Â· 1× fraction 0â€"1 Â· 1× signed â^'1â€¦1 Â· 1× count

top_beta

GENE | REAL â‰¥ 0 | AGG FAMILY | 100%

The largest Î² among the gene's families. Range **0.00096 â€“ 1.52**, 100% filled. AUTHORED

example: `CDH1 â†' 0.2584`

---

top_beta_frac

GENE | FRACTION 0â€"1 | AGG FAMILY | 100%

That family's SHARE of the gene's total Î². Range **0.076 â€“ 1**, 100% filled. AUTHORED

example: `CDH1 â†' 0.1919`

---

top_family_identity

GENE | IDENTIFIER | AGG FAMILY | 87%

The family Shapley/LMG credits with the gene's regulation. NaN for 197 genes (12.7%) where NNLS zeroed everything â€" an honest 'undefined', which the magnitude answer cannot express. AUTHORED

example: `CDH1 â†' miR-25-3p/32-5p/92-3p/363-3p/367-3p`

---

top_family_magnitude

GENE | IDENTIFIER | AGG FAMILY | 100%

The family with the largest beta. AUTHORED

example: `CDH1 â†' miR-25-3p/32-5p/92-3p/363-3p/367-3p`

---

top_gaining_arms

GENE | IDENTIFIER | 91%

Which arms gained the most repression healthyâ†'tumour. Range **1,279 distinct**, 91% filled. AUTHORED

example: `CDH1 â†' hsa-miR-30a-5p(+0.25);hsa-miR-9-5p(+0.24);hsa-miR-544a-3p(+0.02);hsa-miR-34c-3p(+0.01);hsa-miR-25-3p(+0.01);hsa-miR-199a-5p(-0.00)`

---

top_identity

GENE | REAL â‰¥ 0 | AGG FAMILY | 87%

The largest `identity` share among the gene's families. AUTHORED

example: `CDH1 â†' 0.4477`

---

top_identity_gated

GENE | SIGNED â^'1â€¦1 | 78%

`top_identity` recomputed over `identity_reliable` edges only â€" **max 1.0000, 0 genes above 1** (vs +740.007 ungated). Defined on 1,203 genes. AUTHORED

example: `CDH1 â†' 0.3367`

---

top_identity_n_reliable

GENE | COUNT | 78%

How many of the gene's families survive the identity gate â€" the DENOMINATOR behind `top_identity_gated`. **Median 2.** AUTHORED

example: `CDH1 â†' 26`

REALIZED_ Â· 5

supporting â€" read these when the core raises a question

5 columns Â· rung **gene** Â· coverage **63%â€"82%** (median 81%)
2× signed â^'1â€¦1 Â· 1× boolean Â· 1× count Â· 1× real, signed

realized_composition_explained

GENE | BOOLEAN | 82%

Is the gene's REALIZED coupling composition-explained (retention < 0.4)? AUTHORED

values: `False (785) Â· True (490)`

example: `CDH1 â†' True`

---

realized_n_reg
GENE | COUNT | 82%

How many regulators the realized coupling was computed over. Range **1 â€“ 63**, 82% filled. AUTHORED

example: CDH1 â†’ 17

---

realized_retention
GENE | REAL, SIGNED | 63%

Realized-coupling retention, Ï_adjusted / Ï_raw. AUTHORED

example: CDH1 â†’ -0.16

---

realized_rho_adj
GENE | SIGNED Â−1â€¦1 | 81%

The same, composition-adjusted. Range **-0.479 â€“ +0.509**, 81% filled. AUTHORED

example: CDH1 â†’ -0.035

---

realized_rho_raw
GENE | SIGNED Â−1â€¦1 | 81%

The gene's realized coupling, UNADJUSTED. Range **-0.67 â€“ +0.62**, 81% filled. AUTHORED

example: CDH1 â†’ 0.225

W_ Â· 1

supporting â€” read these when the core raises a question

> **1** columns Â· rung **gene** Â· coverage **100%â€“100%** (median 100%)
> 1× real â‰¥ 0

w_max
GENE | REAL â‰¥ 0 | 100%

The MAXIMUM curated evidence weight `w` over the gene's regulators (`gene_atlas` : `nanmax(w)`), where `w` comes from the same PMID-deduped ledger that feeds `lit_` and `fame_` . AUTHORED

example: CDH1 â†’ 13.24

LIT_ Â· 7

reference â€” context, provenance and rarely-quoted detail

> **7** columns Â· rung **gene** Â· coverage **21%â€“21%** (median 21%)
> 4× count Â· 2× boolean Â· 1× identifier

lit_agrees_identity
GENE | BOOLEAN | 21%

Does `identity` name the same family as the **most-published** one for this gene â€” **True on 100 of 323 (31%)**. AUTHORED

values: False (223) Â· True (100)

example: CDH1 â†’ False

---

lit_agrees_magnitude
GENE | BOOLEAN | 21%

Does the model's largest-Î² family match the most-published one (**96 of 329 genes, 29%**)? AUTHORED

values: False (233) Â· True (96)

example: CDH1 â†’ False

---

lit_family
GENE | IDENTIFIER | 21%

The most-published seed family for this gene by distinct low-throughput functional PMIDs. AUTHORED

example: CDH1 â†’ miR-9-5p

---

lit_margin
GENE | COUNT | 21%

Gap between the top and second literature family, in PMIDs â€” **median 1.00**, i.e. the literature's 'winner' is usually decided by a SINGLE paper. â‡’ **gate on this before reading `lit_agrees_*`**: a 1-PMID margin is a coin flip dressed as ground truth. AUTHORED

example: CDH1 â†’ 3

> *Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)*

---

lit_n_lit_families

GENE | COUNT | 21%

â›”â›” **A STUDY-COUNT AXIS, NOT A BIOLOGICAL GROUND TRUTH â€” read the name with suspicion.** `eval/lit_ground_truth.py` defines the 'canonical' family of a gene as the **argmax of DISTINCT PubMed IDs** carrying low-throughput functional evidence (reporter / western / proteomics; miRTarBase-Weak and TarBase-Positive excluded). AUTHORED

example: `CDH1 â†' 25`

---

lit_n_pmid

GENE | COUNT | 21%

Distinct low-throughput functional PMIDs behind `lit_family`. Median **3** â€” the 'ground truth' rests on a handful of papers per gene. AUTHORED

example: `CDH1 â†' 6`

> *Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)*

---

lit_tier

GENE | COUNT | 21%

Literature-evidence tier for the gene (levels 2, 3 and 4; 4 is the most populated at 132 genes). AUTHORED

values: `4.0 (132) Â· 2.0 (129) Â· 3.0 (68)`

example: `CDH1 â†' 4`

CPTAC_ Â· 24

reference â€” context, provenance and rarely-quoted detail

---

24 columns Â· rung **gene** Â· AGGREGATE of the edge block via an agg_/abund_ infix â€” NOT derivable (MH-262) Â· coverage **42%â€“89%** (median 66%)
18× signed âˆ'1â€¦1 Â· 2× boolean Â· 2× real, signed Â· 2× count

cptac_prosp_abund_rho_disc

GENE | SIGNED Âˆ'1Â€¦1 | AGG ARM | 72%

Spearman coupling of an **UNWEIGHTED ABUNDANCE SUM** â€” the reference the aggregate is judged against against the discovery/decoy layer in the PROSPECTIVE cohort (independent patients, TMT-11, nâ‰ˆ101), adjusted for the C block (composition + target CN + proliferation). AUTHORED

example: `CDH1 â†' -0.08033`

---

cptac_prosp_abund_rho_prot

GENE | SIGNED Âˆ'1Â€¦1 | AGG ARM | 73%

Spearman coupling of an **UNWEIGHTED ABUNDANCE SUM** â€” the reference the aggregate is judged against against PROTEIN in the PROSPECTIVE cohort (independent patients, TMT-11, nâ‰ˆ101), adjusted for the C block (composition + target CN + proliferation). AUTHORED

example: `CDH1 â†' 0.109`

---

cptac_prosp_abund_rho_rna

GENE | SIGNED Âˆ'1Â€¦1 | AGG ARM | 74%

Spearman coupling of an **UNWEIGHTED ABUNDANCE SUM** â€” the reference the aggregate is judged against against mRNA in the PROSPECTIVE cohort (independent patients, TMT-11, nâ‰ˆ101), adjusted for the C block (composition + target CN + proliferation). AUTHORED

example: `CDH1 â†' 0.2848`

---

cptac_prosp_agg_beats_abund_prot

GENE | BOOLEAN | 73%

Does the learned Î²-weighted aggregate couple more negatively to PROTEIN than a plain abundance sum, in the CPTAC prospective cohort? AUTHORED

values: `False (770) Â· True (357)`

example: `CDH1 â†' True`

---

cptac_prosp_agg_ret_prot

GENE | REAL, SIGNED | 59%

`rho_prot(composition-adjusted) / rho_prot_raw` â€” the ADJUSTMENT sense of retention, on the prospective cohort. âœ… **The denominator is ALREADY GATED** at `RHO_GATE` in `cptac_card.py:322`, so values outside [0,1] are NOT the vanishing-denominator artifact of axiom 5 (verified: 0 rows with |raw| < 0.02). AUTHORED

example: `CDH1 â†' 0.855`

---

cptac_prosp_agg_rho_disc

GENE | SIGNED Âˆ'1Â€¦1 | AGG ARM | 72%

Spearman coupling of the **Î²-WEIGHTED AGGREGATE** of the gene's arms against the discovery/decoy layer in the PROSPECTIVE cohort (independent patients, TMT-11, nâ‰ˆ101), adjusted for the C block (composition + target CN + proliferation). AUTHORED

example: `CDH1 â†' -0.1539`

cptac_prosp_agg_rho_disc_raw
GENE | SIGNED Â˘'1Â€¦1 | 72%

Spearman coupling of the **Î²-WEIGHTED AGGREGATE** of the gene's arms against the discovery/decoy layer in the PROSPECTIVE cohort (independent patients, TMT-11, nâ‰ˆ101), **UNADJUSTED** â€” no confounder block. AUTHORED
example: CDH1 â†' -0.1971

---

cptac_prosp_agg_rho_prot
GENE | SIGNED Â˘'1Â€¦1 | AGG ARM | 73%

Spearman coupling of the **Î²-WEIGHTED AGGREGATE** of the gene's arms against PROTEIN in the PROSPECTIVE cohort (independent patients, TMT-11, nâ‰ˆ101), adjusted for the C block (composition + target CN + proliferation). AUTHORED
example: CDH1 â†' -0.1211

---

cptac_prosp_agg_rho_prot_raw
GENE | SIGNED Â˘'1Â€¦1 | 73%

Spearman coupling of the **Î²-WEIGHTED AGGREGATE** of the gene's arms against PROTEIN in the PROSPECTIVE cohort (independent patients, TMT-11, nâ‰ˆ101), **UNADJUSTED** â€” no confounder block. AUTHORED
example: CDH1 â†' -0.1416

---

cptac_prosp_agg_rho_rna
GENE | SIGNED Â˘'1Â€¦1 | AGG ARM | 74%

Spearman coupling of the **Î²-WEIGHTED AGGREGATE** of the gene's arms against mRNA in the PROSPECTIVE cohort (independent patients, TMT-11, nâ‰ˆ101), adjusted for the C block (composition + target CN + proliferation). AUTHORED
example: CDH1 â†' 0.04264

---

cptac_prosp_agg_rho_rna_raw
GENE | SIGNED Â˘'1Â€¦1 | 74%

Spearman coupling of the **Î²-WEIGHTED AGGREGATE** of the gene's arms against mRNA in the PROSPECTIVE cohort (independent patients, TMT-11, nâ‰ˆ101), **UNADJUSTED** â€” no confounder block. AUTHORED
example: CDH1 â†' 0.06539

---

cptac_prosp_n_arms
GENE | COUNT | 89%

Arms contributing to this gene's prospective-cohort aggregate (median 2, max 61). AUTHORED
example: CDH1 â†' 23

---

cptac_t105_abund_rho_disc
GENE | SIGNED Â˘'1Â€¦1 | 60%

Spearman coupling of an **UNWEIGHTED ABUNDANCE SUM** â€” the reference the aggregate is judged against against the discovery/decoy layer in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED
example: CDH1 â†' -0.0005806

---

cptac_t105_abund_rho_prot
GENE | SIGNED Â˘'1Â€¦1 | 60%

Spearman coupling of an **UNWEIGHTED ABUNDANCE SUM** â€” the reference the aggregate is judged against against PROTEIN in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED
example: CDH1 â†' 0.01071

---

cptac_t105_abund_rho_rna
GENE | SIGNED Â˘'1Â€¦1 | 60%

Spearman coupling of an **UNWEIGHTED ABUNDANCE SUM** â€” the reference the aggregate is judged against against mRNA in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED
example: CDH1 â†' 0.06946

---

cptac_t105_agg_beats_abund_prot
GENE | BOOLEAN | 60%

As the prospective twin, on the TCGA-105 overlap cohort. AUTHORED
values: False (633) Â· True (299)
example: CDH1 â†' True

---

cptac_t105_agg_ret_prot
GENE | REAL, SIGNED | 42%

As the prospective twin, TCGA-105. AUTHORED
example: CDH1 â†' 0.2276

---

cptac_t105_agg_rho_disc
GENE | SIGNED Â˘'1Â€¦1 | AGG ARM | 60%

Spearman coupling of the **Î²-WEIGHTED AGGREGATE** of the gene's arms against the discovery/decoy layer in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED
example: CDH1 â†' 0.04722

cptac_t105_agg_rho_disc_raw

GENE | SIGNED Â˜'1â€¦1 | 60%

Spearman coupling of the **Î²-WEIGHTED AGGREGATE** of the gene's arms against the discovery/decoy layer in the TCGA-OVERLAP cohort (n=105), **UNADJUSTED** â€” no confounder block. AUTHORED

example: `CDH1 â†' -0.03473`

---

cptac_t105_agg_rho_prot

GENE | SIGNED Â˜'1â€¦1 | AGG ARM | 60%

Spearman coupling of the **Î²-WEIGHTED AGGREGATE** of the gene's arms against PROTEIN in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED

example: `CDH1 â†' -0.02019`

---

cptac_t105_agg_rho_prot_raw

GENE | SIGNED Â˜'1â€¦1 | 60%

Spearman coupling of the **Î²-WEIGHTED AGGREGATE** of the gene's arms against PROTEIN in the TCGA-OVERLAP cohort (n=105), **UNADJUSTED** â€” no confounder block. Range **-0.484 â€“ +0.446**, 60% filled. AUTHORED

example: `CDH1 â†' -0.08874`

---

cptac_t105_agg_rho_rna

GENE | SIGNED Â˜'1â€¦1 | AGG ARM | 60%

Spearman coupling of the **Î²-WEIGHTED AGGREGATE** of the gene's arms against mRNA in the TCGA-OVERLAP cohort (n=105), adjusted for the C block (composition + target CN + proliferation). AUTHORED

example: `CDH1 â†' -0.07787`

---

cptac_t105_agg_rho_rna_raw

GENE | SIGNED Â˜'1â€¦1 | 60%

Spearman coupling of the **Î²-WEIGHTED AGGREGATE** of the gene's arms against mRNA in the TCGA-OVERLAP cohort (n=105), **UNADJUSTED** â€” no confounder block. Range **-0.636 â€“ +0.54**, 60% filled. AUTHORED

example: `CDH1 â†' -0.1085`

---

cptac_t105_n_arms

GENE | COUNT | 85%

Arms behind the TCGA-105 aggregate (median 2, max 57). See the prospective twin. AUTHORED

example: `CDH1 â†' 21`

> *Defined on: genes/edges present in the CPTAC cohort*

FRAC_ Â· 3

reference â€” context, provenance and rarely-quoted detail

> **3** columns Â· rung **gene** Â· coverage **60%â€“100%** (median 82%)
> 3× fraction 0â€“1

frac_edges_realized

GENE | FRACTION 0â€“1 | 82%

Fraction of the gene's scored edges that realize a coupling below the threshold. AUTHORED

example: `CDH1 â†' 0.7059`

---

frac_gain_true_acquired

GENE | FRACTION 0â€“1 | 60%

Fraction of the gene's acquired-repression calls that survive the healthy-leg triage â€” i.e. are not `collapse_blind` artifacts. AUTHORED

example: `CDH1 â†' 0`

---

frac_identified

GENE | FRACTION 0â€“1 | 100%

Fraction of this gene's families reaching `|z| > 2` on the UNCALIBRATED SD. AUTHORED

example: `CDH1 â†' 0.16`

DOMINANT_ Â· 4

reference â€” context, provenance and rarely-quoted detail

> **4** columns Â· rung **gene** Â· coverage **77%â€“83%** (median 82%)
> 3× identifier Â· 1× categorical

dominant_HLY

GENE | IDENTIFIER | 77%

The arm carrying the largest share of the gene's HEALTHY budget. Range **280 distinct**, 77% filled. AUTHORED

example: CDH1 â†’ hsa-miR-92a-3p

---

dominant_NAT

GENE | IDENTIFIER | 81%

The arm dominating its NAT budget. Range **309 distinct**, 81% filled. AUTHORED

example: CDH1 â†’ hsa-miR-30a-5p

---

dominant_TUM

GENE | IDENTIFIER | 82%

The arm carrying the largest share of the gene's tumour pressure budget (miR-21-5p on 56 genes, miR-200c-3p next). AUTHORED

example: CDH1 â†’ hsa-miR-30a-5p

---

dominant_edge_shift_class

GENE | CATEGORICAL | 83%

How the gene's dominant edge moved healthyâ†’tumour. Range **10 levels**, 83% filled. AUTHORED

example: CDH1 â†’ uncoupled

STATIC_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **87%â€“87%** (median 87%)
1× identifier

static_owner_family

GENE | IDENTIFIER | 87%

The static (dose-based) owner of the gene, for comparison against the realization owner. AUTHORED

example: CDH1 â†’ miR-25-3p/32-5p/92-3p/363-3p/367-3p

> *Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)*

TOTAL_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **100%â€“100%** (median 100%)
1× real â‰¥ 0

total_pressure

GENE | REAL Â‰¥ 0 | 100%

Sum over the gene's regulators. AUTHORED

example: CDH1 â†’ 1.353

OWNER_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **41%â€“41%** (median 41%)
1× boolean

owner_agrees

GENE | BOOLEAN | 41%

Whether the realization owner and the static owner agree. Defined only where both exist. AUTHORED

values: False (385) Â· True (253)

example: CDH1 â†’ True

> *Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)*

MEDIAN_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **100%â€“100%** (median 100%)
1× real â‰¥ 0

median_retention
GENE · REAL â‰¥ 0 · 100%

A per-row median of a lower-rung quantity; check `agg_of`. AUTHORED

example: CDH1 â†’ 0.9148

ARMRES_ Â· 7

reference â€” context, provenance and rarely-quoted detail

7 columns Â· rung **gene** Â· coverage **18%â€“18%** (median 18%)
2× signed âˆ’1â€¦1 Â· 2× count Â· 1× categorical Â· 1× fraction 0â€“1 Â· 1× real â‰¥ 0

armres_best_drho
GENE · SIGNED âˆ’1â€¦1 · 18%

The largest arm-resolution gain over the family fit. Range **-0.206 â€“ +0.0421**, 18% filled. AUTHORED

example: CDH1 â†’ -0.004735

armres_class
GENE · CATEGORICAL · 18%

`arm_modelable` (every multi-arm cell resolvable, 68 genes) Â· `partial` (37) Â· `family_only` (no cell resolvable, 168). Absent = the gene has no multi-arm cell at all, so the question is undefined. AUTHORED

values: family_only (168) Â· arm_modelable (68) Â· partial (37)

example: CDH1 â†’ arm_modelable

armres_frac_resolvable
GENE · FRACTION 0â€“1 · 18%

Fraction of this gene's multi-arm cells where the arms are separable. AUTHORED

example: CDH1 â†’ 1

armres_med_drho
GENE · SIGNED âˆ’1â€¦1 · 18%

Median `oof_drho` over the gene's multi-arm cells â€” negative means arm resolution predicts repression better out of fold. AUTHORED

example: CDH1 â†’ -0.004735

armres_med_sep_z
GENE · REAL â‰¥ 0 · 18%

Median separation z between same-cell arms. Range **0.0536 â€“ 18.5**, 18% filled. AUTHORED

example: CDH1 â†’ 3.196

armres_n_arms_split
GENE · COUNT · 18%

How many arms sit in those multi-arm cells. Range **2 â€“ 40**, 18% filled. AUTHORED

example: CDH1 â†’ 2

armres_n_multi_cells
GENE · COUNT · 18%

How many of the gene's family cells hold >1 arm. Range **1 â€“ 14**, 18% filled. AUTHORED

example: CDH1 â†’ 1

IDENTITY_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **87%â€“87%** (median 87%)
1× boolean

identity_eq_magnitude
GENE · BOOLEAN · AGG FAMILY · 87%

Do the identity and magnitude answers name the SAME top family. Exactly 100% agreement at n_fam==1 by construction; 76% at n_fam>=2. Read it only on multi-family genes. AUTHORED

values: True (1,161) Â· False (191)

example: CDH1 â†’ True

GREAL_ Â· 10

reference â€” context, provenance and rarely-quoted detail

10 columns Â· rung **gene** Â· coverage **29%â€“83%** (median 83%)
7× count Â· 2× signed âˆ’1â€¦1 Â· 1× fraction 0â€“1

greal_best_coupling
GENE | SIGNED Âˆ'1Â€¦1 | 83%

The STRONGEST (most negative) realized coupling over this gene's scored regulators. AUTHORED

example: `CDH1 â†' -0.271`

greal_frac_c10
GENE | FRACTION 0Â€“1 | 29%

Gene-rung twin of `real_frac_c10` â€” defined on 446 genes against `greal_n_c10`'s 1,289; gated-out genes have a median `greal_n_scored` of **1**. Exact on 100%. AUTHORED

example: `CDH1 â†' 0.4`

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€” the GENE-side ladder; rates gated nâ‰¥3*

greal_med_coupling
GENE | SIGNED Âˆ'1Â€¦1 | 83%

The MEDIAN realized coupling over this gene's scored regulators. Range **-0.503 â€“ +0.298**, 83% filled. AUTHORED

example: `CDH1 â†' -0.083`

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€” the GENE-side ladder; rates gated nâ‰¥3*

greal_n_c05
GENE | COUNT | 83%

How many of this gene's scored regulators couple **below âˆ'0.05**. AUTHORED

example: `CDH1 â†' 7`

greal_n_c10
GENE | COUNT | 83%

How many of this gene's scored regulators couple **below âˆ'0.10**. AUTHORED

example: `CDH1 â†' 4`

greal_n_c15
GENE | COUNT | 83%

How many of this gene's scored regulators couple **below âˆ'0.15**. AUTHORED

example: `CDH1 â†' 2`

greal_n_c20
GENE | COUNT | 83%

How many of this gene's scored regulators couple **below âˆ'0.20**. AUTHORED

example: `CDH1 â†' 2`

greal_n_c30
GENE | COUNT | 83%

How many of this gene's scored regulators couple **below âˆ'0.30**. AUTHORED

example: `CDH1 â†' 0`

greal_n_scored
GENE | COUNT | 83%

Genes/edges scored for realized coupling (median 2, max 59) â€” the DENOMINATOR of every `greal_frac_*`. âœ… The `greal_n_c05 â€¦ n_c30` ladder is properly nested: **0 monotonicity violations and 0 rows where a threshold count exceeds `n_scored`** (verified). AUTHORED

example: `CDH1 â†' 10`

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€” the GENE-side ladder; rates gated nâ‰¥3*

greal_n_sig
GENE | COUNT | 83%

How many of this gene's scored regulators reach significance. AUTHORED

example: `CDH1 â†' 2`

HEUR_ Â· 5

reference â€” context, provenance and rarely-quoted detail

5 columns Â· rung **gene** Â· coverage **89%â€“91%** (median 91%)
2× signed âˆ'1â€¦1 Â· 1× count Â· 1× boolean Â· 1× categorical

heur_delta_tumor_nat
GENE | SIGNED Âˆ'1Â€¦1 | 89%

Tumour-minus-NAT change in the heuristic pressure correlation. AUTHORED

example: `CDH1 â†' 0.03725`

heur_n_regulators

GENE | COUNT | 91%

âš âš **A DIFFERENT PIPELINE'S COUNT â€" not the model's.** It is `edges.groupby('gene')['miRNA'].nunique()` inside `analyses/misc/mirna_comovement.py`, i.e. AUTHORED

example: `CDH1 â†' 12`

---

heur_net_repressed_tumor

GENE | BOOLEAN | 91%

Is the gene net-repressed in tumour, per the RETIRED heuristic lane. True for 282 of 1,409 genes. AUTHORED

values: `False (1,127) Â· True (282)`

example: `CDH1 â†' False`

---

heur_repression_class

GENE | CATEGORICAL | 91%

Cross-state repression class from the RETIRED heuristic lane (never_repressed / lost_repression / gained_repression / constitutive_repressed). 867 of 1,409 genes are `never_repressed`. AUTHORED

values: `never_repressed (867) Â· lost_repression (258) Â· gained_repression (167) Â· constitutive_repressed (115) Â· na (2)`

example: `CDH1 â†' never_repressed`

---

heur_rho_pressure_tumor

GENE | SIGNED Âˆ'1Â€¦1 | 91%

Spearman of the gene's HEURISTIC aggregate pressure against its mRNA in tumour. AUTHORED

example: `CDH1 â†' 0.05168`

ACQUIRED_VS_ Â· 2

reference â€" context, provenance and rarely-quoted detail

2 columns Â· rung **gene** Â· coverage **91%â€"91%** (median 91%)
2× real, signed

acquired_vs_gtex

GENE | REAL, SIGNED | 91%

Repression GAINED relative to the GTEx healthy baseline. Range **-11.6 â€" 7.56**, 91% filled. AUTHORED

example: `CDH1 â†' 0.04213`

---

acquired_vs_nat

GENE | REAL, SIGNED | 91%

Repression gained relative to matched NAT â€" the same-platform comparison, and the one to prefer. Range **-4.24 â€" 2.83**, 91% filled. AUTHORED

example: `CDH1 â†' 0.3754`

PRESSURE_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **91%â€"91%** (median 91%)
1× real â‰¥ 0

pressure_tumor

GENE | REAL Â‰¥ 0 | 91%

Aggregate incoming pressure on the gene in the named state. AUTHORED

example: `CDH1 â†' 3.784`

TCGA_ Â· 4

reference â€" context, provenance and rarely-quoted detail

4 columns Â· rung **gene** Â· coverage **89%â€"89%** (median 89%)
3× signed âˆ'1â€¦1 Â· 1× count

tcga_abund_rho_rna

GENE | SIGNED Âˆ'1Â€¦1 | AGG ARM | 89%

TCGA mRNA coupling of an UNWEIGHTED abundance sum â€" the reference Î² is judged against. Range **-0.36 â€" +0.366**, 89% filled. AUTHORED

example: `CDH1 â†' 0.006381`

tcga_agg_rho_rna

GENE · SIGNED Âˆ'1â€¦1 · AGG ARM · 89%

Gene-level Î²-weighted aggregate coupling to mRNA in TCGA (median **âˆ'0.05**) against `tcga_abund_rho_rna`'s **âˆ'0.02** â€" the abundance-sum reference. AUTHORED

example: `CDH1 â†' -0.1922`

---

tcga_agg_rho_rna_raw

GENE · SIGNED Âˆ'1â€¦1 · 89%

The Î²-weighted aggregate mRNA coupling, UNADJUSTED. Range **-0.545 â€" +0.571**, 89% filled. AUTHORED

example: `CDH1 â†' -0.1921`

---

tcga_n_arms

GENE · COUNT · 89%

Arms behind the TCGA aggregate. Range **1 â€" 62**, 89% filled. AUTHORED

example: `CDH1 â†' 23`

REALIZATION_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **42%â€"42%** (median 42%)
1× identifier

realization_owner

GENE · IDENTIFIER · 42%

Within-patient realization quantities â€" the score, its identity allocation, and who owns it. AUTHORED

example: `CDH1 â†' miR-25-3p/32-5p/92-3p/363-3p/367-3p`

> *Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)*

DISC_ Â· 2

reference â€" context, provenance and rarely-quoted detail

2 columns Â· rung **gene** Â· coverage **3%â€"100%** (median 51%)
1× categorical Â· 1× count

disc_gold_families

GENE · CATEGORICAL · 3%

Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed). AUTHORED

example: `KLF3 â†' miR-130-3p/301-3p/454-3p;miR-17-5p/20-5p/93-5p/106-5p/519-3p`

---

disc_n_gold_edges

GENE · COUNT · 100%

How many of the 157 discovery-queue edges name this gene (max 4). AUTHORED

values: `0 (1,510) Â· 1 (26) Â· 3 (5) Â· 2 (5) Â· 4 (3)`

example: `CDH1 â†' 0`

MAX_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **100%â€"100%** (median 100%)
1× fraction 0â€"1

max_beta_frac_sd

GENE · FRACTION 0â€"1 · 100%

Maximum over the row's constituent units. AUTHORED

example: `CDH1 â†' 0.03738`

REGULATORY_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **100%â€"100%** (median 100%)
1× boolean

regulatory_handoff
GENE BOOLEAN 100%

Does the gene's globally-dominant regulator differ between healthy and tumour OR between NAT and tumour. AUTHORED

values: False (1,221) Â· True (328)

example: CDH1 â†’ True

MEAN_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **82%â€“82%** (median 82%)
1× real, signed

mean_dTarget
GENE REAL, SIGNED 82%

A per-row mean of a lower-rung quantity; check `agg_of` for which rung it summarises. AUTHORED

example: CDH1 â†’ 1.18

CONCENTRATION_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **53%â€“53%** (median 53%)
1× fraction 0â€“1

concentration_adj
GENE FRACTION 0â€“1 53%

`(concentration âˆ’ 1/n_fam) / (1 âˆ’ 1/n_fam)` âˆˆ [0,1] â€” how concentrated the gene is RELATIVE to the most diffuse its own design allows. AUTHORED

example: CDH1 â†’ 0.1573

FAM_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **81%â€“81%** (median 81%)
1× signed âˆ’1â€¦1

fam_pooled_rho_adj
GENE SIGNED âˆ’1â€¦1 81%

Seed-family structure the arm sits in â€” membership, affinity and dominance within it. AUTHORED

example: CDH1 â†’ -0.096

gene_family_card.tsv

gene_family

One row per one (seed family, gene) pair. Keyed [gene, family] â€" it cannot express a property of the family itself. That is the seed_family card's job.

61 columns **5,117** rows **12** blocks median coverage **97%**

Contents â€" 12 blocks, most important first

(BARE) Â· 8

core â€" the columns findings rest on

8 columns Â· **spans family, key** â€" this prefix groups columns living on DIFFERENT units Â· coverage **95%â€"100%** (median 100%)
3× identifier Â· 3× real â‰¥ 0 Â· 1× boolean Â· 1× real, signed

arms
FAMILY IDENTIFIER 100%

The arms collapsed into this (gene, family) cell, **semicolon-separated** (`hsa-miR-17-5p;hsa-miR-20a-5p`). AUTHORED
example: `CDH1 / miR-132-3p/212-3p â†' hsa-miR-132-3p`

beta
FAMILY REAL â‰¥ 0 100%

FAMILY-rung coupling coefficient â€" `readouts.run(level='family')` , the same-seed collapse APPLIED. AUTHORED
example: `CDH1 / miR-132-3p/212-3p â†' 0.07163`

family
KEY IDENTIFIER 100%

KEY â€" seed family of the regulator, as used by the family-rung fit. AUTHORED
example: `CDH1 â†' miR-132-3p/212-3p`

gene
KEY IDENTIFIER 100%

KEY â€" HGNC gene symbol. AUTHORED
example: `miR-9-3p â†' ITGB1`

identified
FAMILY BOOLEAN 100%

`|z| > 2` on the UNCALIBRATED SD â€" **27.1%** of cells. `cal_identified` is the honest version. AUTHORED
values: False (3,730) Â· True (1,387)
example: `CDH1 / miR-132-3p/212-3p â†' False`

identity
FAMILY REAL, SIGNED 95%

Shapley/LMG identity at the FAMILY rung â€" same estimator as the edge card's `identity` , different unit. AUTHORED
example: `CDH1 / miR-132-3p/212-3p â†' 0.0641`

retention
FAMILY REAL â‰¥ 0 100%

`beta_deconv / beta_core` at the family grain â€" the ADJUSTMENT sense of retention (what fraction of the coupling survives the composition block), never the patient-baseline sense. AUTHORED
example: `CDH1 / miR-132-3p/212-3p â†' 0.8516`

z
FAMILY | REAL â‰¥ 0 | 100%

`beta / beta_sd` for the family cell, on the **UNCALIBRATED** posterior SD. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 1.844`

N_ Â· 3

supporting â€” read these when the core raises a question

> **3** columns Â· **spans family, gene** â€” this prefix groups columns living on DIFFERENT units Â· coverage **100%â€“100%** (median 100%)
> 2× count Â· 1× constant

n_arms
FAMILY | COUNT | 100%

How many arms of THIS family sit in THIS gene's design. âœ… FAMILY rung, VERIFIED (2026-08-19): it varies within gene for 241 of 1,549 genes, and the per-gene SUM of these equals the gene card's `n_arms` exactly â€” a clean cross-rung consistency check. AUTHORED

values: `1 (4,646) Â· 2 (338) Â· 3 (87) Â· 4 (25) Â· 5 (11) Â· 6 (6) Â· 7 (4)`

example: `CDH1 / miR-132-3p/212-3p â†’ 1`

n_fam
GENE | COUNT | 100%

How many families compete at this gene. beta is mathematically INERT at 1. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 25`

n_samples
GENE | CONSTANT | 100%

Samples the family cell was fit on â€” **CONSTANT 1,040** across the card. AUTHORED

values: `1040 (5,117)`

example: `CDH1 / miR-132-3p/212-3p â†’ 1040`

CTX_ Â· 16

supporting â€” read these when the core raises a question

> **16** columns Â· rung **gene** Â· SUBSET of the edge block, identical rungs Â· coverage **81%â€“100%** (median 96%)
> 4× signed âˆ’1â€¦1 Â· 3× real â‰¥ 0 Â· 3× real, signed Â· 3× count Â· 1× categorical Â· 1× fraction 0â€“1 Â· 1× boolean

ctx_apriori_class
GENE | CATEGORICAL | 100%

The a-priori measurability class assigned to the gene's design before any fit (`gene_atlas` via `card_context`). Range **4 levels**, 100% filled. AUTHORED

values: `A_COMPETENT (3,402) Â· D_UNDEFINED (730) Â· B_PARTIAL (590) Â· C_WEAK (395)`

example: `CDH1 / miR-132-3p/212-3p â†’ A_COMPETENT`

ctx_ceiling
GENE | SIGNED Â−1â€¦1 | 99%

The UPPER BOUND on what ANY estimator can achieve for this gene â€” the 5-fold out-of-fold RÂ² of an UNRESTRICTED OLS on the gene's real design (`card_context.py:15`). AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0.1535`

ctx_collapse
GENE | REAL â‰¥ 0 | 96%

`n_arm / n_fam` for the gene â€” **how much curation BUNDLES same-seed mates onto it** (1â€“5). AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 1.091`

ctx_d_collin
GENE | SIGNED Â−1â€¦1 | 81%

Mean pairwise COLLINEARITY among the gene's regulators â€” how much they move together (`gene_atlas` via `card_context`). Range **-0.699 â€“ +0.495**, 81% filled. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ -0.02497`

ctx_d_fam
GENE | REAL, SIGNED | 95%

Design width at FAMILY grain (`gene_atlas` via `card_context`). Range **-2 â€“ 25**, 95% filled. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0`

ctx_dose_max

GENE | REAL Â‰¥ 0 | 96%

The LARGEST regulator dose in the gene's design (`gene_atlas` via `card_context`). Range **0.215 â€“ 17.8**, 96% filled. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 15.38`

---

ctx_dose_med

GENE | REAL Â‰¥ 0 | 96%

The MEDIAN regulator dose in the gene's design (`gene_atlas` via `card_context`). Range **0.215 â€“ 17.8**, 96% filled. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 6.318`

---

ctx_frac_abund

GENE | FRACTION 0Â€“1 | 96%

FRACTION of the gene's regulators clearing the abundance gate (`gene_atlas` via `card_context`). Range **0 â€“ 1**, 96% filled. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0.5833`

---

ctx_gap_core

GENE | SIGNED Â ˆ’1Â€¦1 | 95%

The Î²-vs-decoy MARGIN under the CORE confounder block (`gene_atlas` via `card_context`). Range **-0.476 â€“ +0.541**, 95% filled. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ -0.2714`

---

ctx_gap_d_dose

GENE | REAL, SIGNED | 95%

Difference in dose between the real design and its decoy. Range **-5.41 â€“ 1.74**, 95% filled. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0.1238`

---

ctx_gap_deconv

GENE | SIGNED Â ˆ’1Â€¦1 | 95%

The same margin under the DECONVOLUTION block (`gene_atlas` via `card_context`). Range **-0.33 â€“ +0.374**, 95% filled. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ -0.1864`

---

ctx_gap_retention

GENE | REAL, SIGNED | 92%

`gap_deconv / gap_core` â€” how much of the decoy margin survives adjustment (`gene_atlas` via `card_context`). Range **-16.5 â€“ 17**, 92% filled. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0.6867`

---

ctx_measurable

GENE | BOOLEAN | 99%

Is the gene's `ctx_ceiling` above `CEILING_ROBUST` (**0.02**, not 0 â€” MH-144)? AUTHORED

values: `True (3,652) Â· False (1,432)`

example: `CDH1 / miR-132-3p/212-3p â†’ True`

---

ctx_n_abund

GENE | COUNT | 96%

How many of the gene's regulators clear the abundance gate (`gene_atlas` via `card_context`). Range **0 â€“ 53**, 96% filled. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 7`

---

ctx_n_fam_abund

GENE | COUNT | 96%

How many families clear the abundance gate. Range **0 â€“ 34**, 96% filled. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 6`

---

ctx_n_fam_multi

GENE | COUNT | 96%

How many of the gene's families hold MORE THAN ONE arm (`gene_atlas` via `card_context`). Range **0 â€“ 14**, 96% filled. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 1`

BETA_ Â· 7

supporting â€” read these when the core raises a question

> **7** columns Â· rung **family** Â· coverage **100%â€“100%** (median 100%)
> 4× fraction 0â€“1 Â· 3× real â‰¥ 0

beta_deconv

FAMILY | REAL Â‰¥ 0 | 100%

The same Î² refit under the DECONVOLUTION confounder block. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0.061`

---

beta_frac

FAMILY | FRACTION 0Â€“1 | 100%

Î²_f / Î£Î² averaged over Gibbs DRAWS â€” E[ratio]. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0.05285`

beta_frac_abs

FAMILY | FRACTION 0â€“1 | 100%

|E[Î²_f]| / Î£|E[Î²]| â€” the share from POINT ESTIMATES. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0.05296`

beta_frac_deconv

FAMILY | FRACTION 0â€“1 | 100%

The share recomputed under the deconvolution block. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0.05107`

beta_frac_sd

FAMILY | FRACTION 0â€“1 | 100%

Posterior SD of the per-draw share. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0.02825`

beta_sd

FAMILY | REAL â‰¥ 0 | 100%

Posterior SD of Î² on the UNCALIBRATED scale. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0.03885`

beta_sd_deconv

FAMILY | REAL â‰¥ 0 | 100%

Posterior SD of the deconv-block Î². AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0.03579`

COMP_ Â· 16

reference â€” context, provenance and rarely-quoted detail

16 columns Â· rung **gene** Â· coverage **66%â€“97%** (median 81%)
6× categorical Â· 6× signed âˆ'1â€¦1 Â· 3× real, signed Â· 1× real â‰¥ 0

comp_cptac_prot_driver

GENE | CATEGORICAL | 66%

WHICH deconvolved compartment most explains this gene's PROTEIN coupling â€” `prolif` 219 Â· `CAFs` 173 Â· `purity` 140 Â· `PVL` 77 Â· `Endothelial` 67. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ T-cells`

comp_cptac_prot_driver_ret

GENE | REAL, SIGNED | 66%

`1 âˆ' comp_cptac_prot_driver_share`, exactly. Median +0.398 raw, **+0.469** gated to |Ï_raw| â‰¥ 0.10 â€” quote the gated value. See the mRNA twin. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0.5085`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

comp_cptac_prot_driver_share

GENE | REAL, SIGNED | 66%

Protein-layer twin of `comp_tcga_mrna_driver_share`. Same exact complementarity with `_driver_ret` (8.9e-16 on 918 rows). AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0.4915`

comp_cptac_prot_rho_raw

GENE | SIGNED âˆ'1â€¦1 | 81%

Unadjusted aggregate PROTEIN coupling â€” denominator of `comp_cptac_prot_driver_ret`. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ -0.1416`

comp_cptac_prot_top_budget

GENE | CATEGORICAL | 81%

The compartment the budget loads on most (protein). Range **10 levels**, 81% filled. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ Normal Epithelial`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

comp_cptac_prot_top_budget_r

GENE | SIGNED âˆ'1â€¦1 | 81%

As the TCGA mRNA twin, on CPTAC protein â€” the budget-vs-compartment correlation. Median **âˆ'0.118**, i.e. it runs the OPPOSITE way to mRNA (+0.121). AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0.3126`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

comp_cptac_prot_top_target

GENE | CATEGORICAL | 81%

The compartment the target loads on most (protein). Range **10 levels**, 81% filled. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ T-cells`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

comp_cptac_prot_top_target_r

GENE | SIGNED Â˘'1â€¦1 | 81%

As the TCGA mRNA twin, on CPTAC protein â€" the target-vs-compartment correlation. Median **+0.203**. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' -0.2385`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_tcga_mrna_driver

GENE | CATEGORICAL | 81%

WHICH compartment drives the gene's mRNA coupling â€" the one whose removal moves it most (`CAFs` 279 Â· `purity` 174 Â· `prolif` 148 Â· `Endothelial` 122 Â· `Myeloid` 109). AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' T-cells`

---

comp_tcga_mrna_driver_ret

GENE | REAL, SIGNED | 81%

`1 â˘' comp_tcga_mrna_driver_share`, exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.8681`

---

comp_tcga_mrna_driver_share

GENE | REAL â‰¥ 0 | 81%

Share of the gene's mRNA coupling attributable to the single compartment that most changes it â€" **axiom 8's composition GATE**, the axis that separates the harm tails at p=0.00999 where the ensemble axis gives p=0.10. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.1319`

---

comp_tcga_mrna_rho_raw

GENE | SIGNED Â˘'1â€¦1 | 97%

The gene's UNADJUSTED aggregate mRNA coupling â€" the DENOMINATOR of `comp_tcga_mrna_driver_ret`. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' -0.1921`

---

comp_tcga_mrna_top_budget

GENE | CATEGORICAL | 97%

The compartment the gene's miRNA BUDGET loads on most â€" a property of the REGULATORS. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' Normal Epithelial`

---

comp_tcga_mrna_top_budget_r

GENE | SIGNED Â˘'1â€¦1 | 97%

Correlation between the gene's **miRNA PRESSURE BUDGET** and the fraction of the compartment that budget loads on most (`comp_tcga_mrna_top_budget`), in TCGA mRNA. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.2545`

---

comp_tcga_mrna_top_target

GENE | CATEGORICAL | 97%

The compartment the TARGET GENE's expression loads on most. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' prolif`

---

comp_tcga_mrna_top_target_r

GENE | SIGNED Â˘'1â€¦1 | 97%

Correlation between the **TARGET GENE's own expression** and the fraction of the compartment it loads on most. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.3289`

IDENTITY_ Â· 4

reference â€" context, provenance and rarely-quoted detail

> **4** columns Â· rung **family** Â· coverage **95%â€"97%** (median 95%)
> 2× fraction 0â€"1 Â· 1× real, signed Â· 1× boolean

identity_abs

FAMILY | FRACTION 0â€"1 | 95%

|identity| / Î£|identity| per gene â€" the BOUNDED companion to `identity`, always in [0,1] (verified: 0 non-NaN values outside it on either card). AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.063`

---

identity_coherence

FAMILY | FRACTION 0â€"1 | 95%

1 / Î£|identity| per GENE, in (0,1] â€" identity's own sign coherence. 1.0 when nothing cancels, â†' 0 as suppressor cancellation grows. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.983`

identity_deconv

FAMILY | REAL, SIGNED | 97%

Shapley/LMG attribution on R^2 with bagged-NNLS weights â€" WHO regulates the gene, collinearity-fair. AUTHORED

example: CDH1 / miR-132-3p/212-3p â†' 0.04796

identity_reliable

FAMILY | BOOLEAN | 95%

`identity_coherence â‰¥ 0.5` AND `|identity| â‰¤ 1`. Admits **93.3%** of edge rows and **92.6%** of gene_family rows, and excludes **every** row whose |identity| exceeds 1 (119/119 and 118/118). AUTHORED

values: True (4,740) Â· False (146)

example: CDH1 / miR-132-3p/212-3p â†' True

PIP_ Â· 2

reference â€" context, provenance and rarely-quoted detail

2 columns Â· rung **family** Â· coverage **100%â€"100%** (median 100%)
1× constant Â· 1× fraction 0â€"1

pip_dense

FAMILY | CONSTANT | 100%

âš CONSTANT by construction (Ï€â‰¡1 dense readout). AUTHORED

values: 1.0 (5,117)

example: CDH1 / miR-132-3p/212-3p â†' 1

pip_discovery

FAMILY | FRACTION 0â€"1 | 100%

Posterior inclusion probability. `pip_dense` comes from the pi=1 coupling readout; `pip_discovery` from the evidence-pi readout. AUTHORED

example: CDH1 / miR-132-3p/212-3p â†' 0.3261

W_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **100%â€"100%** (median 100%)
1× real â‰¥ 0

w_max

GENE | REAL â‰¥ 0 | 100%

Maximum curated EVIDENCE weight for this (gene, family) cell. Family-rung here; gene-rung on the gene card. AUTHORED

example: CDH1 / miR-132-3p/212-3p â†' 13.24

M_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **family** Â· coverage **100%â€"100%** (median 100%)
1× real â‰¥ 0

m_nnls

FAMILY | REAL â‰¥ 0 | 100%

Bagged-NNLS weight for this (gene, family) cell. AUTHORED

example: CDH1 / miR-132-3p/212-3p â†' 0.07211

COMPOSITION_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **family** Â· coverage **100%â€"100%** (median 100%)
1× categorical

composition_class

FAMILY CATEGORICAL 100%

Composition-adjustment verdict for this row (e.g. `composition_explained` when the effect does not survive the deconvolution block). AUTHORED

values: cell_intrinsic (3,378) Â· partial (1,095) Â· composition_explained (644)

example: CDH1 / miR-132-3p/212-3p â†’ cell_intrinsic

PRIOR_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **family** Â· coverage **100%â€"100%** (median 100%)
1× fraction 0â€"1

prior_pi

FAMILY FRACTION 0â€"1 100%

The evidence prior fed to the discovery readout. AUTHORED

example: CDH1 / miR-132-3p/212-3p â†’ 0.2962

RETENTION_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **family** Â· coverage **100%â€"100%** (median 100%)
1× boolean

retention_reliable

FAMILY BOOLEAN 100%

Fraction of an effect surviving covariate adjustment (adjusted / raw). AUTHORED

values: False (3,730) Â· True (1,387)

example: CDH1 / miR-132-3p/212-3p â†’ False arm_card.tsv

arm

One row per the miRNA arm itself, gene-free. The deepest card. An arm-rung column must be constant within arm ACROSS genes; if it varies by gene it is not an arm property, whatever its name says.

296 columns **2,450** rows **44** blocks median coverage **30%**

Contents â€” 44 blocks, most important first

ADM_ Â· 4

core â€” the columns findings rest on

4 columns Â· rung **arm** Â· AGGREGATE of the edge block â€” verified: n_with_site and n_admissible reproduce a direct rollup on 100% of arms (n_edges only 77.5%, the MH-252 universe gap) Â· coverage **14%â€“29%** (median 29%)
3× count Â· 1× fraction 0â€“1

adm_frac_with_site
ARM FRACTION 0â€“1 14%

`adm_n_with_site / adm_n_edges`. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.9429

adm_n_admissible
ARM COUNT 29%

Edges of this arm that pass admissibility (`adm_has_site` + the evidence bar) â€” MH-216's single most load-bearing conditioning variable, rolled up per arm. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 33

adm_n_edges
ARM | COUNT | 29%

ADMISSIBILITY â€” could this edge repress in breast AT ALL? `has_site` (seed match in the 3'UTR) AND `expressed` (arm present in TCGA tumour OR GTEx healthy breast). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 35`

---

adm_n_with_site
ARM | COUNT | 29%

Edges of this arm whose target carries a seed site â€” **89.9% of edges (4,436/4,937)**, so the site requirement is NOT the binding constraint; the evidence bar is (admissible drops it to 60.4%). 19.7% of arms have none. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 33`

> *Defined on: edges/arms with an admissibility tag (4,937 edges; has_site 89.9% Â· expressed 63.9% Â· admissible 60.4%)*

ARM_ Â· 7

core â€” the columns findings rest on

7 columns Â· rung **arm** Â· coverage **15%â€"20%** (median 20%)
4× real, signed Â· 2× real â‰¥ 0 Â· 1× percent

arm_iqr
ARM | REAL â‰¥ 0 | 20%

Interquartile range of the arm's tumour RPM â€” its DYNAMIC RANGE. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 1.1`

---

arm_lfc_HLY_NAT_raw
ARM | REAL, SIGNED | 15%

log-FC GTEx-healthy â†' TCGA-NAT, raw. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 7.94`

---

arm_lfc_HLY_TUM_QN
ARM | REAL, SIGNED | 16%

log-FC GTEx-healthy â†' TCGA-tumour, on quantile-normalised values. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 5.11`

---

arm_lfc_HLY_TUM_raw
ARM | REAL, SIGNED | 15%

As `arm_lfc_HLY_TUM_QN` but against the RAW GTEx median. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 10`

---

arm_lfc_NAT_TUM
ARM | REAL, SIGNED | 20%

log-FC of the arm's median abundance, matched NAT â†' tumour. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 2.06`

---

arm_med_rpm
ARM | REAL â‰¥ 0 | 20%

The arm's MEDIAN RPM across tumour samples â€” its abundance level, gene-free. Defined on the 19.8% of arms with tumour expression data. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 13.4`

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

---

arm_pct_above_floor
ARM | PERCENT | 20%

â›" **THE NAME READS BACKWARDS.** Computed as `100Â·mean(x > FLOOR)` â€" the percent of samples ABOVE the detection floor, i.e. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 100`

SEED_ Â· 1

core â€” the columns findings rest on

1 columns Â· rung **arm** Â· coverage **100%â€"100%** (median 100%)
1× identifier

seed_family
ARM | IDENTIFIER | 100%

KEY â€” the seed family itself, one row per family. AUTHORED

example: `hsa-let-7a-5p â†' let-7-5p/98-5p`

ARB_ Â· 16

supporting â€” read these when the core raises a question

> **16** columns Â· rung **arm** Â· coverage **28%â€“30%** (median 30%)
> 12× count Â· 2× real â‰¥ 0 Â· 1× fraction 0â€“1 Â· 1× signed â´’1â€¦1

arb_frac_identified
ARM | FRACTION 0â€“1 | AGG EDGE | 30%

`arb_n_identified / arb_n_edges` (verified exact on 100% of rows). AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.1316

---

arb_max_abs_beta
ARM | REAL â‰¥ 0 | AGG EDGE | 30%

Largest |Î²| among the arm's edges. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.2173

---

arb_max_identity
ARM | SIGNED â´’1â€¦1 | AGG EDGE | 28%

Largest `identity` among this arm's edges, over reliable rows only. âœ… Now runs **â´’0.255 â€¦ +1.000** â€” it reached **+740.0** until the inert `beta_frac_reliable` gate was replaced (2026-08-19). AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1

---

arb_mean_abs_beta
ARM | REAL â‰¥ 0 | AGG EDGE | 30%

Mean |Î²| over the arm's edges. See `arb_max_abs_beta` for the one-edge degeneracy (9.3% of arms, where the two are equal by construction). AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.02837

> *Defined on: arms with a learned-Î² edge rollup (728)*

---

arb_n_b001
ARM | COUNT | 30%

How many of this arm's edges carry **Î² â‰¥ 0.001** . AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 18

---

arb_n_b001_ident
ARM | COUNT | 30%

How many of this arm's edges carry **Î² â‰¥ 0.001** AND identified (|z|>2). AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 5

---

arb_n_b005
ARM | COUNT | 30%

How many of this arm's edges carry **Î² â‰¥ 0.005** . AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 6

---

arb_n_b005_ident
ARM | COUNT | 30%

How many of this arm's edges carry **Î² â‰¥ 0.005** AND identified (|z|>2). AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 5

---

arb_n_b010
ARM | COUNT | 30%

How many of this arm's edges carry **Î² â‰¥ 0.010** . AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 4

---

arb_n_b010_ident
ARM | COUNT | 30%

How many of this arm's edges carry **Î² â‰¥ 0.010** AND identified (|z|>2). AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 4

---

arb_n_b025
ARM | COUNT | 30%

How many of this arm's edges carry **Î² â‰¥ 0.025** . AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0

---

arb_n_b025_ident
ARM | COUNT | 30%

How many of this arm's edges carry **Î² â‰¥ 0.025** AND identified (|z|>2). AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0

---

arb_n_comp_explained
ARM | COUNT | AGG EDGE | 30%

How many of this arm's edges are classed `composition_explained` . AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0

---

arb_n_edges
ARM | COUNT | AGG EDGE | 30%

Edges this arm carries in the LEARNED MODEL's own readouts. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 38

---

arb_n_identified
ARM | COUNT | AGG EDGE | 30%

How many of this arm's edges reach significance. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 5

arb_n_identity_reliable

ARM COUNT AGG EDGE 30%

Edges of this arm whose `identity` share is usable (`identity_reliable`). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 33

HEALTHY_ Â· 3

supporting â€" read these when the core raises a question

3 columns Â· rung **arm** Â· identical column set to the edge block Â· coverage **16%â€"20%** (median 16%)
1× categorical Â· 1× real â‰¥ 0 Â· 1× boolean

healthy_leg

ARM CATEGORICAL 20%

Provenance of this arm's GTEx healthy leg â€" `measured`, `collapse_blind` or `true_absent`. AUTHORED

values: measured (230) Â· true_absent (168) Â· collapse_blind (100)

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' measured

healthy_potential

ARM REAL â‰¥ 0 16%

The arm's IMPUTED healthy abundance â€" a repression-POTENTIAL proxy, collapse-repaired (0.27 â€" 17.9). AUTHORED

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 8.24

healthy_uninformative

ARM BOOLEAN 16%

Is the healthy leg both `collapse_blind` AND high-potential â€" i.e. the one combination where an *acquired* call is genuinely unsafe? **True on 143 of 3,490 edges (4%).** AUTHORED

values: False (392) Â· True (7)

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' False

SITE_ Â· 12

supporting â€" read these when the core raises a question

12 columns Â· rung **arm** Â· coverage **29%â€"29%** (median 29%)
7× count Â· 2× fraction 0â€"1 Â· 2× real, signed Â· 1× real â‰¥ 0

site_frac_8mer

ARM FRACTION 0â€"1 29%

8mer share of the arm's **CANONICAL** sites (`site_n_8mer / site_n_canonical`; median 0.150). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.2169

site_frac_canonical

ARM FRACTION 0â€"1 29%

Canonical share of the arm's **TOTAL** sites (`site_n_canonical / site_n_total`; median 0.054) â€" a different denominator from `site_frac_8mer`. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.03913

site_n_6mer

ARM COUNT AGG EDGE 29%

Count of 6mer sites for this arm in the **MANE 3'UTR seed scan**, scoped to Hallmark genes. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1459

site_n_7mer

ARM COUNT AGG EDGE 29%

Count of 7mer sites for this arm in the **MANE 3'UTR seed scan**, scoped to Hallmark genes. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 278

site_n_8mer

ARM COUNT AGG EDGE 29%

Count of 8mer sites for this arm in the **MANE 3'UTR seed scan**, scoped to Hallmark genes. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 77

site_n_canonical

ARM COUNT AGG EDGE 29%

Count of CANONICAL sites (6/7/8mer) for this arm in the **MANE 3'UTR seed scan**, scoped to Hallmark genes. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 355

site_n_genes_canonical
ARM | COUNT | AGG EDGE | 29%

Count of GENES carrying a canonical site for this arm in the **MANE 3'UTR seed scan**, scoped to Hallmark genes. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 283`

---

site_n_noncanon
ARM | COUNT | AGG EDGE | 29%

Count of NON-canonical sites for this arm in the **MANE 3'UTR seed scan**, scoped to Hallmark genes. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 7258`

---

site_n_total
ARM | COUNT | AGG EDGE | 29%

Count of ALL sites for this arm in the **MANE 3'UTR seed scan**, scoped to Hallmark genes. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 9072`

---

site_repression_med
ARM | REAL, SIGNED | AGG EDGE | 29%

Median predicted repression score over the arm's sites (all negative by construction; median âˆ'0.316). A PREDICTION from sequence, not a measurement â€" never cite it as evidence of observed repression. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' -0.3859`

> *Defined on: arms in the scanMiR site-type table (771) â€" âš HALLMARK-SCOPED, 1,432 genes only*

---

site_repression_min
ARM | REAL, SIGNED | AGG EDGE | 29%

Strongest (most negative) predicted repression among the arm's sites (median âˆ'1.964). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' -2.437`

---

site_sites_per_gene
ARM | REAL Â‰¥ 0 | 29%

Canonical sites per targeted gene (median 1.34) â€" site DENSITY, separating an arm with many sites on few genes from one spread thin. `site_n_canonical / site_n_genes_canonical`. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 1.254`

> *Defined on: arms in the scanMiR site-type table (771) â€" âš HALLMARK-SCOPED, 1,432 genes only*

FAME_ Â· 28

supporting â€" read these when the core raises a question

28 columns Â· rung **arm** Â· coverage **94%â€"96%** (median 96%)
27× count Â· 1× fraction 0â€"1

fame_assay_binding__functional_mti_weak_studies
ARM | COUNT | 96%

Curated studies for this arm that are BOTH binding assays (CLIP, pull-down) AND carry weak functional MTI evidence â€" a CELL of the assay Ã— functional-class cross-tab. Range **0 â€" 928**, 96% of arms. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 150`

---

fame_assay_binding_studies
ARM | COUNT | 96%

Studies whose assay class is BINDING (the marginal of the assay Ã— functional-class cross-tab). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 150`

---

fame_assay_functional_mti_studies
ARM | COUNT | 96%

Curated studies for this arm using functional_mti assays â€" a MARGINAL of the assay Ã— functional-class cross-tab, so it is the sum of that row's cells. Range **0 â€" 70**, 96% of arms. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 35`

---

fame_assay_functional_mti_weak_studies
ARM | COUNT | 96%

Curated studies for this arm using functional_mti_weak assays â€" a MARGINAL of the assay Ã— functional-class cross-tab, so it is the sum of that row's cells. Range **0 â€" 929**, 96% of arms. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 158`

fame_assay_nonfunctional_mti_studies ARM COUNT 96%

Curated studies for this arm using nonfunctional_mti assays â€” a MARGINAL of the assay Ã— functional-class cross-tab, so it is the sum of that row's cells. Range **0 â€“ 35**, 96% of arms. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 13`

---

fame_assay_other__functional_mti_studies ARM COUNT 96%

Curated studies for this arm that are BOTH assays outside the four named classes AND carry strong functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 4`

---

fame_assay_other__functional_mti_weak_studies ARM COUNT 96%

Curated studies for this arm that are BOTH assays outside the four named classes AND carry weak functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 2`

---

fame_assay_other__nonfunctional_mti_studies ARM COUNT 96%

Curated studies for this arm that are BOTH assays outside the four named classes AND carry explicitly NON-functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED

values: `0.0 (2,225) Â· 1.0 (77) Â· 2.0 (17) Â· 3.0 (10) Â· 5.0 (6) Â· 4.0 (4) Â· 6.0 (3) Â· 7.0 (1)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 1`

---

fame_assay_other_studies ARM COUNT 96%

Curated studies for this arm using assays outside the four named classes â€” a MARGINAL of the assay Ã— functional-class cross-tab, so it is the sum of that row's cells. Range **0 â€“ 259**, 96% of arms. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 6`

---

fame_assay_protein__functional_mti_studies ARM COUNT 96%

Curated studies for this arm that are BOTH protein-level assays (western, ELISA) AND carry strong functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 28`

---

fame_assay_protein__functional_mti_weak_studies ARM COUNT 96%

Curated studies for this arm that are BOTH protein-level assays (western, ELISA) AND carry weak functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED

values: `0.0 (2,225) Â· 1.0 (75) Â· 2.0 (22) Â· 3.0 (10) Â· 4.0 (6) Â· 5.0 (3) Â· 10.0 (1) Â· 25.0 (1)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0`

---

fame_assay_protein__nonfunctional_mti_studies ARM COUNT 96%

Curated studies for this arm that are BOTH protein-level assays (western, ELISA) AND carry explicitly NON-functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 10`

---

fame_assay_protein_studies ARM COUNT 96%

Curated studies for this arm using protein-level assays (western, ELISA) â€” a MARGINAL of the assay Ã— functional-class cross-tab, so it is the sum of that row's cells. Range **0 â€“ 61**, 96% of arms. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 31`

---

fame_assay_reporter__functional_mti_studies ARM COUNT 96%

Curated studies for this arm that are BOTH reporter assays (luciferase) AND carry strong functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. Range **0 â€“ 65**, 96% of arms. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 31`

`fame_assay_reporter__nonfunctional_mti_studies`

ARM COUNT 96%

Curated studies for this arm that are BOTH reporter assays (luciferase) AND carry explicitly NON-functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 10`

`fame_assay_reporter_studies`

ARM COUNT 96%

Curated studies for this arm using reporter assays (luciferase) â€” a MARGINAL of the assay Ã— functional-class cross-tab, so it is the sum of that row's cells. Range **0 â€“ 73**, 96% of arms. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 34`

`fame_assay_rna__functional_mti_studies`

ARM COUNT 96%

Curated studies for this arm that are BOTH RNA-level assays (qPCR, microarray) AND carry strong functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 22`

`fame_assay_rna__functional_mti_weak_studies`

ARM COUNT 96%

Curated studies for this arm that are BOTH RNA-level assays (qPCR, microarray) AND carry weak functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 9`

`fame_assay_rna__nonfunctional_mti_studies`

ARM COUNT 96%

Curated studies for this arm that are BOTH RNA-level assays (qPCR, microarray) AND carry explicitly NON-functional MTI evidence â€” a CELL of the assay Ã— functional-class cross-tab. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 5`

`fame_assay_rna_studies`

ARM COUNT 96%

Curated studies for this arm using RNA-level assays (qPCR, microarray) â€” a MARGINAL of the assay Ã— functional-class cross-tab, so it is the sum of that row's cells. Range **0 â€“ 638**, 96% of arms. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 30`

`fame_assay_total_studies`

ARM COUNT 96%

Distinct STUDIES (`study_id.nunique()`) behind this arm's curated edges â€” median 57, max 930. See `fame_assay_unique_targets` : different definitions, one axis (spearman 1.0000). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 189`

`fame_assay_unique_targets`

ARM COUNT 96%

Distinct TARGET GENES this arm has in the curated literature (`gene.nunique()`). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 189`

`fame_led_frac_ltfunc`

ARM FRACTION 0â€“1 AGG EDGE 96%

Their share of the arm's publications â€” **median 0.00**. Range **0 â€“ 1**, 96% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.4565`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)*

`fame_led_n_classes`

ARM COUNT AGG EDGE 96%

How many distinct evidence CLASSES this arm's curated edges span (1â€“7). AUTHORED

values: `1.0 (1,466) Â· 5.0 (246) Â· 6.0 (241) Â· 4.0 (166) Â· 2.0 (137) Â· 3.0 (76) Â· 7.0 (18)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 6`

`fame_led_n_ltfunc_pmid`

ARM COUNT AGG EDGE 96%

Low-throughput FUNCTIONAL publications â€” the strongest rung. Range **0 â€“ 197**, 96% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 42`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)*

`fame_led_n_weak`

ARM COUNT AGG EDGE 94%

Weak-evidence publications for this arm. Range **1 â€“ 754**, 94% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 206`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)*

fame_n_genes_curated
ARM | COUNT | AGG EDGE | 96%

Distinct genes this arm has in the curated literature. Range **1 â€“ 3615**, 96% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 3216`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)*

fame_npmid
ARM | COUNT | AGG EDGE | 96%

Distinct PMIDs citing this arm. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 92`

(BARE) Â· 1

supporting â€” read these when the core raises a question

> **1** columns Â· rung **key** Â· coverage **100%â€“100%** (median 100%)
> 1× identifier

arm
KEY | IDENTIFIER | 100%

KEY â€” mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through `_arm_key()`, which gates against the canonical universe and LOGS drops rather than silently dropping. AUTHORED

example: `miR-588 â†’ hsa-miR-4701-5p`

HLY_ Â· 7

supporting â€” read these when the core raises a question

> **7** columns Â· rung **arm** Â· coverage **17%â€“91%** (median 91%)
> 2× real â‰¥ 0 Â· 1× categorical Â· 1× boolean Â· 1× percent Â· 1× count Â· 1× fraction 0â€“1

hly_baseline_src
ARM | CATEGORICAL | 91%

WHERE the healthy baseline came from â€” `floor0` (1,137) / `mited_anchor` (634) / `gtex` (417) / `gtex_family` (49). Range **4 levels**, 91% filled. AUTHORED

values: `floor0 (1,137) Â· mited_anchor (634) Â· gtex (417) Â· gtex_family (49)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ mited_anchor`

> *Defined on: arms with a GTEx healthy anchor â€” âš MEASURED leg only; floor0 is NEVER emitted as a value*

hly_from_seedmate
ARM | BOOLEAN | 91%

Was this arm's healthy baseline borrowed from a SEED-MATE rather than measured directly (49 arms)? AUTHORED

values: `False (2,188) Â· True (49)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ False`

hly_gtex_median
ARM | REAL â‰¥ 0 | 17%

The arm's median GTEx healthy abundance. Range **0.0803 â€“ 18.1**, 17% filled. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†’ 11.62`

> *Defined on: arms with a GTEx healthy anchor â€” âš MEASURED leg only; floor0 is NEVER emitted as a value*

hly_gtex_pct
ARM | PERCENT | 17%

Its GTEx percentile. Range **0.0514 â€“ 1**, 17% filled. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†’ 0.9636`

> *Defined on: arms with a GTEx healthy anchor â€” âš MEASURED leg only; floor0 is NEVER emitted as a value*

hly_nat_detect_n
ARM | COUNT | 91%

In how many of the ~104 matched NAT samples this arm is detected above the floor. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 104`

hly_nat_median
ARM | REAL â‰¥ 0 | 74%

Median NAT abundance (fill 74.0%). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 15.09`

hly_tumor_prev
ARM | FRACTION 0â€“1 | 91%

Prevalence of the arm above the detection floor in tumour (median 0.01). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 1`

DOSE_ Â· 3

supporting â€” read these when the core raises a question

3 columns Â· rung **arm** Â· SUBSET of the edge block Â· coverage **12%â€“12%** (median 12%)
2× real â‰¥ 0 Â· 1× constant

dose_comp_retention
ARM | REAL â‰¥ 0 | 12%

Arm-rung version of the composition dose retention. Range **0.51 â€“ 1.66**, 12% filled. AUTHORED

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†’ 0.99

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

dose_confounded
ARM | CONSTANT | 12%

Is this arm's NATâ†’tumour dose a composition or proliferation artifact (either gated retention < 0.4)? AUTHORED

values: False (291)

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†’ False

dose_prolif_retention
ARM | REAL â‰¥ 0 | 12%

Arm-rung version of the proliferation dose retention. Range **0.69 â€“ 1.25**, 12% filled. AUTHORED

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†’ 1

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

AGO_ Â· 4

supporting â€” read these when the core raises a question

4 columns Â· rung **arm** Â· âš arm.ago_dom and edge.ago_loading are RANK-IDENTICAL (spearman +1.000, equal on 73.2%) â€” one quantity, two names. Never scan both. Â· coverage **28%â€“60%** (median 28%)
1× fraction 0â€“1 Â· 1× constant Â· 1× categorical Â· 1× real â‰¥ 0

ago_dom
ARM | FRACTION 0â€“1 | 28%

Fraction of the arm's Manakov AGO-CLIP reads assigned to it rather than its sibling arm â€” the loading dominance. Median 0.658 on the 27.7% of arms with CLIP coverage. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.9937

ago_dom_src
ARM | CONSTANT | 28%

Provenance of `ago_dom` â€” which CLIP dataset the AGO-loading dominance came from. AUTHORED

values: manakov (678)

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ manakov

ago_guide_class
ARM | CATEGORICAL | 28%

AGO/RISC loading gate for the arm. AUTHORED

values: guide (288) Â· co_loaded (204) Â· passenger (186)

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ guide

ago_reads
ARM | REAL â‰¥ 0 | 60%

Raw AGO-CLIP read count (median 6, max 210,722). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 9900

ABUND_ Â· 5

supporting â€” read these when the core raises a question

5 columns Â· rung **arm** Â· coverage **88%â€“91%** (median 91%)
3× real â‰¥ 0 Â· 1× fraction 0â€“1 Â· 1× categorical

abund_frac_detected
ARM | FRACTION 0â€“1 | 91%

Fraction of samples where the arm clears the floor. Range **0 â€“ 1**, 91% filled. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1

> *Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)*

abund_iqr
ARM | REAL â‰¥ 0 | 91%

Its interquartile range. Range **0 â€“ 4.59**, 91% filled. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.9309

> *Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)*

---

abund_med
ARM | REAL â‰¥ 0 | 91%

The arm's median abundance across tumour samples. Range **0.105 â€“ 17.9**, 91% filled. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 14.8

> *Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)*

---

abund_sd
ARM | REAL â‰¥ 0 | 88%

Its dispersion â€" AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.7103

---

abund_sd_tertile
ARM | CATEGORICAL | 88%

Dispersion tertile â€" **718 / 716 / 718 by construction**, so it carries no information beyond `abund_sd` 's rank. AUTHORED

values: high (718) Â· low (718) Â· mid (716)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' high

ATTR_ Â· 14

supporting â€" read these when the core raises a question

> **14** columns Â· rung **arm** Â· coverage **6%â€"29%** (median 15%)
> 11× fraction 0â€"1 Â· 2× count Â· 1× signed âˆ'1â€¦1

attr_max_budget_share
ARM | FRACTION 0â€"1 | 15%

Largest value of the arm's share of its gene's DOSE budget over this arm's cells. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.2604

---

attr_max_credit_share
ARM | FRACTION 0â€"1 | 6%

Largest value of the arm's share of attribution CREDIT over this arm's cells. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.8519

---

attr_max_joint_share
ARM | FRACTION 0â€"1 | 15%

Largest JOINT attribution share among this arm's edges. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.3167

---

attr_max_nested_share
ARM | FRACTION 0â€"1 | 15%

Largest NESTED attribution share. See `attr_max_joint_share` : different values (agree on 35.4%), one axis (spearman +0.962). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.3167

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

---

attr_max_within_share
ARM | FRACTION 0â€"1 | 15%

Largest WITHIN-cell attribution share for this arm. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1

---

attr_med_budget_share
ARM | FRACTION 0â€"1 | 15%

Median value of the arm's share of its gene's DOSE budget over this arm's cells. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.1036

---

attr_med_credit_share
ARM | FRACTION 0â€"1 | 6%

Median value of the arm's share of attribution CREDIT over this arm's cells. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.2493

---

attr_med_dose_share
ARM | FRACTION 0â€"1 | 29%

Median dose share of this arm within its family cells. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.1582

---

attr_med_joint_share
ARM | FRACTION 0â€"1 | 15%

Median value of the JOINT (Shapley) share over this arm's cells. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.07153

---

attr_med_nested_share
ARM | FRACTION 0â€"1 | 15%

Median value of the NESTED share over this arm's cells. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.1393

attr_med_phi_joint
ARM | SIGNED Âˆ'1Â€¦1 | 15%

Median value of the joint Shapley Ï† over this arm's cells. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.008336`

---

attr_med_within_share
ARM | FRACTION 0Â€"1 | 15%

Median value of the WITHIN-cell share over this arm's cells. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.33`

---

attr_n_edges
ARM | COUNT | 22%

Attribution rollup over the arm's edges â€" how much of the gene's coupling this arm is credited with, and over how many reliable edges. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 23`

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

---

attr_n_reliable
ARM | COUNT | 15%

Reliable edges behind this arm's `attr_*` shares (median 2). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 4`

CTX_ARM_ Â· 2

reference â€" context, provenance and rarely-quoted detail

---

2 columns Â· rung **arm** Â· the two ARM-RUNG members of ctx_, which the other cards do not carry Â· coverage **29%â€"29%** (median 29%)
1× boolean Â· 1× real â‰¥ 0

ctx_arm_abundant
ARM | BOOLEAN | 29%

Is `ctx_arm_dose >= log2(11)`, i.e. is this arm abundant enough to plausibly repress? **Median 0.00 â€" most arms are NOT.** AUTHORED

values: `False (499) Â· True (213)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' True`

---

ctx_arm_dose
ARM | REAL Â‰¥ 0 | 29%

This arm's median **log2(RPM+1)** across tumour samples (0.13 â€" 17.92, median 1.08). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 14.8`

REAL_ Â· 11

reference â€" context, provenance and rarely-quoted detail

---

11 columns Â· rung **arm** Â· coverage **9%â€"24%** (median 24%)
7× count Â· 2× signed âˆ'1â€¦1 Â· 2× fraction 0â€"1

real_best_coupling
ARM | SIGNED Âˆ'1Â€¦1 | 24%

Most negative realized coupling among this arm's scored genes (median âˆ'0.08, min âˆ'0.56). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' -0.306`

---

real_frac_c10
ARM | FRACTION 0Â€"1 | 9%

Fraction of this arm's scored genes coupling below âˆ'0.10. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.2609`

---

real_frac_sig
ARM | FRACTION 0Â€"1 | 9%

Fraction of this arm's scored genes reaching significance. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.2609`

---

real_med_coupling
ARM | SIGNED Âˆ'1Â€¦1 | 24%

The MEDIAN realized coupling over this arm's scored genes. Range **-0.418 â€" +0.222**, 24% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.046`

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€" the realization LADDER; rates are gated at nâ‰¥5*

---

real_n_c05
ARM | COUNT | 24%

How many of this arm's scored genes couple **below âˆ'0.05**. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 6`

---

real_n_c10
ARM | COUNT | 24%

How many of this arm's scored genes couple **below â^'0.10**. AUTHORED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 6`

---

real_n_c15
ARM | COUNT | 24%

How many of this arm's scored genes couple **below â^'0.15**. AUTHORED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 6`

---

real_n_c20
ARM | COUNT | 24%

How many of this arm's scored genes couple **below â^'0.20**. AUTHORED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 6`

---

real_n_c30
ARM | COUNT | 24%

How many of this arm's scored genes couple **below â^'0.30**. AUTHORED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 1`

---

real_n_scored
ARM | COUNT | 24%

Genes scored for this arm's realized coupling (median 3). âœ... Its `n_c05 â€¦ n_c30` ladder is properly nested â€" **0 violations** (verified). AUTHORED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 23`

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€" the realization LADDER; rates are gated at nâ‰¥5*

---

real_n_sig
ARM | COUNT | 24%

How many of this arm's scored genes reach significance. AUTHORED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 6`

GRANK_ Â· 3

reference â€" context, provenance and rarely-quoted detail

3 columns Â· rung **arm** Â· coverage **15%â€"20%** (median 20%)
3× count

grank_HLY
ARM | COUNT | 15%

The arm's GLOBAL abundance percentile across the whole cohort in the GTEx-healthy state. AUTHORED
example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 96`

---

grank_NAT
ARM | COUNT | 20%

The arm's GLOBAL abundance percentile across the whole cohort in matched normal (NAT). AUTHORED
example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 99`

---

grank_TUM
ARM | COUNT | 20%

The arm's GLOBAL abundance percentile across the whole cohort in tumour. AUTHORED
example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 99`

TS_ Â· 10

reference â€" context, provenance and rarely-quoted detail

10 columns Â· rung **arm** Â· coverage **13%â€"13%** (median 13%)
5× count Â· 2× real, signed Â· 1× signed â^'1â€¦1 Â· 1× fraction 0â€"1 Â· 1× real â‰¥ 0

ts_ctx_best
ARM | REAL, SIGNED | 13%

Best (most negative) TargetScan context++ score for this arm â€" range [â^'2.02, â^'0.34], **all negative by construction** (context++ scores repression as a negative number). AUTHORED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' -0.809`

---

ts_ctx_mean
ARM | SIGNED â^'1â€¦1 | 13%

TargetScan **context++** scan: mean context++ score. AUTHORED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' -0.283`

---

ts_frac_8mer
ARM | FRACTION 0â€"1 | 13%

TargetScan **context++** scan: the 8mer share of this arm's sites. AUTHORED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.3189`

ts_kd_med

ARM | REAL, SIGNED | 13%

TargetScan **context**++ scan: median predicted K_D. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ -5.181

---

ts_n_7mer_A1

ARM | COUNT | 13%

TargetScan **context**++ scan: 7mer-A1 sites. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 463

---

ts_n_7mer_m8

ARM | COUNT | 13%

TargetScan **context**++ scan: 7mer-m8 sites. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 481

---

ts_n_8mer

ARM | COUNT | 13%

TargetScan **context**++ scan: 8mer sites. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 442

---

ts_n_genes

ARM | COUNT | AGG EDGE | 13%

Genes with a TargetScan context++ site â€" median 497, fill **12.9%, the SPARSEST universe on the card**. Correlates +0.726 with the MANE seed scan and only **+0.231** with scanMiR-any. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1207

---

ts_n_sites

ARM | COUNT | AGG EDGE | 13%

TargetScan **context**++ scan: all sites. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1386

---

ts_sites_per_gene

ARM | REAL â‰¥ 0 | 13%

TargetScan **context**++ scan: sites per targeted gene. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1.148

ISOC_ Â· 5

reference â€" context, provenance and rarely-quoted detail

5 columns Â· rung **arm** Â· coverage **91%â€"91%** (median 91%)
3× fraction 0â€"1 Â· 1× real â‰¥ 0 Â· 1× count

isoc_canonical_frac

ARM | FRACTION 0â€"1 | 91%

Fraction of the arm's isomiR reads matching the canonical sequence. Range **0.151 â€" 1**, 91% filled. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.9956

> *Defined on: arms in the isomiR seed-COMPOSITION table â€" where the reads go; â isoc_orphan_frac flags MH-153 mass loss*

---

isoc_donated_frac

ARM | FRACTION 0â€"1 | 91%

Fraction reassigned to a different family. Range **0 â€" 0.5**, 91% filled. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.0029

> *Defined on: arms in the isomiR seed-COMPOSITION table â€" where the reads go; â isoc_orphan_frac flags MH-153 mass loss*

---

isoc_med_reads

ARM | REAL â‰¥ 0 | 91%

Median isomiR reads for this arm â€" **median 1.00, max 528,927**. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 2.229e+04

---

isoc_n_dest_families

ARM | COUNT | 91%

How many families received reads from this arm. Range **0 â€" 6**, 91% filled. AUTHORED

values: 0.0 (1,283) Â· 1.0 (623) Â· 2.0 (238) Â· 3.0 (59) Â· 4.0 (13) Â· 5.0 (3) Â· 6.0 (3)

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1

> *Defined on: arms in the isomiR seed-COMPOSITION table â€" where the reads go; â isoc_orphan_frac flags MH-153 mass loss*

---

isoc_orphan_frac

ARM | FRACTION 0â€"1 | 91%

Fraction belonging to no known family. Range **0 â€" 0.823**, 91% filled. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.0034

> *Defined on: arms in the isomiR seed-COMPOSITION table â€" where the reads go; â isoc_orphan_frac flags MH-153 mass loss*

DGLOBAL_ Â· 3

reference â€" context, provenance and rarely-quoted detail

3 columns Â· rung **arm** Â· coverage **15%â€"20%** (median 15%)
3× real, signed

dGlobal_HLY_NAT
ARM | REAL, SIGNED | 15%

Change in the arm's GLOBAL (cohort-wide) abundance percentile from the GTEx-healthy state to matched normal (NAT). AUTHORED

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 2

dGlobal_HLY_TUM
ARM | REAL, SIGNED | 15%

Change in the arm's GLOBAL (cohort-wide) abundance percentile from the GTEx-healthy state to tumour. AUTHORED

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 3

dGlobal_NAT_TUM
ARM | REAL, SIGNED | 20%

Change in the arm's GLOBAL (cohort-wide) abundance percentile from matched normal (NAT) to tumour. AUTHORED

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 1

CHIM_ Â· 5

reference â€" context, provenance and rarely-quoted detail

5 columns Â· rung **arm** Â· coverage **21%â€"63%** (median 60%)
4× count Â· 1× real â‰¥ 0

chim_manakov_n_genes
ARM | COUNT | AGG EDGE | 60%

Genes with a Manakov chimeric duplex for this arm. Range **0 â€" 13825**, 60% filled. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 7297

> *Defined on: arms with chimeric ligation evidence (1,538) â€" âš ~90% Manakov/HEK293T, per-source, NEVER pooled*

chim_manakov_weight
ARM | REAL â‰¥ 0 | AGG EDGE | 60%

Manakov chimeric support summed over this arm's edges (median 6, max 210,722). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 9900

chim_n_sources
ARM | COUNT | AGG EDGE | 63%

How many chimeric sources support this arm. Range **1 â€" 4**, 63% filled. AUTHORED

values: 1.0 (1,079) Â· 2.0 (229) Â· 4.0 (128) Â· 3.0 (102)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 4

> *Defined on: arms with chimeric ligation evidence (1,538) â€" âš ~90% Manakov/HEK293T, per-source, NEVER pooled*

chim_tarbase_n_genes
ARM | COUNT | AGG EDGE | 21%

Genes with a TarBase chimeric duplex. Range **1 â€" 2701**, 21% filled. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 2701

> *Defined on: arms with chimeric ligation evidence (1,538) â€" âš ~90% Manakov/HEK293T, per-source, NEVER pooled*

chim_tarbase_weight
ARM | COUNT | AGG EDGE | 21%

TarBase chimeric support for the arm (median 8, max 10,255), defined on 20.9% of arms â€" a third of the Manakov coverage. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1.026e+04

> *Defined on: arms with chimeric ligation evidence (1,538) â€" âš ~90% Manakov/HEK293T, per-source, NEVER pooled*

FST_ Â· 16

reference â€” context, provenance and rarely-quoted detail

> **16** columns Â· rung **arm** Â· coverage **3%â€“100%** (median 20%)
> 6× count Â· 5× boolean Â· 3× fraction 0â€“1 Â· 2× signed âˆ’1â€¦1

fst_d_share_HLY_TUM
ARM · SIGNED âˆ’1â€¦1 · 3%

As `fst_d_share_NAT_TUM` but GTEx-healthy â†' tumour, same common-support correction. n=68 (HLY is the sparsest leg at 17.0%). Post-fix **mean âˆ'0.0000, 47.1% positive, p=0.93**. AUTHORED

example: `hsa-miR-26b-5p / miR-26-5p â†' 0.2402`

---

fst_d_share_NAT_TUM
ARM · SIGNED âˆ'1â€¦1 · 6%

Change in the arm's share of its seed family's linear dose, matched NAT â†' tumour, **re-normalised over the members measured in BOTH states**. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' -0.0072`

---

fst_dominance_switch
ARM · BOOLEAN · 7%

Did the family's DOMINANT arm change between healthy and tumour (103 true / 74 false)? AUTHORED

values: `1.0 (103) Â· 0.0 (74)`

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 0`

---

fst_hly_family_measured
ARM · BOOLEAN · 100%

Do ALL members of this arm's family have a measured GTEx level. AUTHORED

values: `False (2,142) Â· True (308)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' False`

---

fst_hly_measured
ARM · BOOLEAN · 100%

Does this arm have a MEASURED healthy leg for the family-share comparison. AUTHORED

values: `False (2,033) Â· True (417)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' False`

---

fst_is_dominant_HLY
ARM · BOOLEAN · 17%

Is the arm its family's dominant member in the healthy state. AUTHORED

values: `True (345) Â· False (72)`

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' True`

---

fst_is_dominant_TUM
ARM · BOOLEAN · 20%

Is it dominant in tumour. AUTHORED

values: `True (394) Â· False (91)`

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' True`

---

fst_n_common_HLY_TUM
ARM · COUNT · 9%

Members measured in BOTH GTEx-healthy and tumour â€" the denominator behind `fst_d_share_HLY_TUM`. See `fst_n_common_NAT_TUM`. AUTHORED

values: `1.0 (162) Â· 2.0 (48) Â· 4.0 (8) Â· 6.0 (6) Â· 3.0 (6)`

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 1`

---

fst_n_common_NAT_TUM
ARM · COUNT · 20%

How many of the family's arms are measured in BOTH NAT and tumour â€" the DENOMINATOR behind `fst_d_share_NAT_TUM`, shipped because the column was wrong for want of it. AUTHORED

values: `1.0 (343) Â· 2.0 (86) Â· 4.0 (16) Â· 3.0 (12) Â· 8.0 (8) Â· 7.0 (7) Â· 6.0 (6)`

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 2`

---

fst_n_members
ARM · COUNT · 100%

How many arms are in this arm's seed family. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 10`

---

fst_rank_HLY
ARM · COUNT · 17%

The arm's rank within its family by healthy dose. AUTHORED

values: `1.0 (348) Â· 2.0 (47) Â· 3.0 (13) Â· 4.0 (6) Â· 5.0 (2) Â· 6.0 (1)`

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 1`

---

fst_rank_NAT
ARM · COUNT · 74%

Its rank by NAT dose. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 1`

fst_rank_TUM
ARM | COUNT | 20%

Its rank by tumour dose. AUTHORED
values: 1.0 (412) Â· 2.0 (46) Â· 3.0 (11) Â· 4.0 (9) Â· 7.0 (2) Â· 5.0 (2) Â· 6.0 (2) Â· 8.0 (1)
example: hsa-miR-200c-3p / miR-200bc-3p/429 â†’ 1

---

fst_share_HLY
ARM | FRACTION 0â€“1 | 17%

Share of family dose in GTEx-healthy, from `hly_gtex_median` (fill **17.0%** â€” the sparsest leg). Same singleton degeneracy; and cross-platform, so contrast only. AUTHORED
example: hsa-miR-200c-3p / miR-200bc-3p/429 â†’ 0.9533

---

fst_share_NAT
ARM | FRACTION 0â€“1 | 74%

Share of family dose in matched NAT, from `hly_nat_median` (fill **74.0%** â€” the best-covered leg, 3.7Ã— TUM). Same singleton degeneracy: median 1.000, mask on `fst_n_members > 1`. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.4124

---

fst_share_TUM
ARM | FRACTION 0â€“1 | 20%

The arm's fraction of its seed family's total linear dose in tumour, from `arm_med_rpm` (fill **19.8%**). AUTHORED
example: hsa-miR-200c-3p / miR-200bc-3p/429 â†’ 0.945

SDSZ_ Â· 4

reference â€” context, provenance and rarely-quoted detail

> **4** columns Â· rung **arm** Â· coverage **97%â€“97%** (median 97%)
> 2× count Â· 1× real â‰¥ 0 Â· 1× fraction 0â€“1

sdsz_N_7mer_plus
ARM | COUNT | AGG EDGE | 97%

7mer-and-better sites for this seed. Range **102 â€“ 8814**, 97% filled. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 3013

> *Defined on: arms in the MANE 3'UTR seed scan â€” âš a FOURTH targetome universe, never blend (2,370)*

---

sdsz_N_8mer
ARM | COUNT | AGG EDGE | 97%

8mer sites for this seed across the scanned universe. Range **8 â€“ 3030**, 97% filled. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 713

> *Defined on: arms in the MANE 3'UTR seed scan â€” âš a FOURTH targetome universe, never blend (2,370)*

---

sdsz_N_eff
ARM | REAL â‰¥ 0 | AGG EDGE | 97%

Effective seed-site count over the scanned universe (median 2,094, fill **96.7%** â€” the best-covered targetome axis). AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1863

---

sdsz_rarity
ARM | FRACTION 0â€“1 | 97%

The arm's seed RARITY, min-max normalised to 0â€“1 (median 0.11). AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.1253

SUB_ Â· 6

reference â€” context, provenance and rarely-quoted detail

> **6** columns Â· rung **arm** Â· coverage **30%â€“30%** (median 30%)
> 2× count Â· 2× signed âˆ’1â€¦1 Â· 1× categorical Â· 1× fraction 0â€“1

sub_best_frac
ARM | FRACTION 0â€“1 | 30%

Subtype rollup: fraction of the arm's edges whose best subtype is the modal one. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.3684

---

sub_best_mode
ARM | CATEGORICAL | 30%

Subtype rollup: the arm's most common best-subtype. AUTHORED
values: Basal (278) Â· Her2 (240) Â· LumB (105) Â· LumA (105)
example: hsa-let-7a-5p / let-7-5p/98-5p â†’ Her2

---

sub_med_rho_best
ARM SIGNED Âˆ'1Â€¦1 30%

Median coupling in the arm's BEST subtype. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' -0.03409`

---

sub_med_rho_pool
ARM SIGNED Âˆ'1Â€¦1 30%

Median pooled-subtype coupling (median âˆ'0.0064) â€" the unselected reference for `sub_med_rho_best`. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.06516`

> *Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) â€" âš DISTRIBUTIONAL (MH-165), not a validated label*

---

sub_n_edges
ARM COUNT 30%

Subtype rollup: edges behind the arm's subtype summary. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 38`

---

sub_n_q05
ARM COUNT 30%

Subtype rollup: how many of the arm's edges reach q<0.05 in ANY subtype. AUTHORED

values: `0.0 (659) Â· 1.0 (54) Â· 2.0 (10) Â· 3.0 (3) Â· 5.0 (2)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0`

COV_ Â· 22

reference â€" context, provenance and rarely-quoted detail

> **22** columns Â· rung **arm** Â· coverage **100%â€"100%** (median 100%)
> 22× boolean

cov_ago_dom
ARM BOOLEAN 100%

Was this arm ever SCANNED for AGO-loading dominance from Manakov CLIP? **True on 678 of 2,450 arms.** AUTHORED

values: `False (1,772) Â· True (678)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' True`

---

cov_attr
ARM BOOLEAN 100%

Was this arm ever SCANNED for the Shapley/NNLS attribution block? **True on 546 of 2,450 arms.** AUTHORED

values: `False (1,904) Â· True (546)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' True`

---

cov_beta
ARM BOOLEAN 100%

Was this arm scanned by the learned-Î² block. Range **728 true / 1,722 false**, 100% filled. AUTHORED

values: `False (1,722) Â· True (728)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' True`

---

cov_cnv
ARM BOOLEAN 100%

Was this arm ever SCANNED for the arm-locus copy-number block? **True on 856 of 2,450 arms.** AUTHORED

values: `False (1,594) Â· True (856)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' True`

---

cov_comp
ARM BOOLEAN 100%

Was this arm ever SCANNED for the compartment-loading (`cload_`) block? **True on 205 of 2,450 arms.** AUTHORED

values: `False (2,245) Â· True (205)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' True`

---

cov_cptac
ARM BOOLEAN 100%

Was it scanned by the CPTAC protein block. Range **2,071 true / 379 false**, 100% filled. AUTHORED

values: `True (2,071) Â· False (379)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' True`

> *Defined on: arms measurable in CPTAC (2,095)*

cov_curated

ARM BOOLEAN 100%

Was this arm ever SCANNED for the curated-literature blocks (`fame_`, `cur_`)? **True on 875 of 2,450 arms.** AUTHORED

values: `False (1,575) Â· True (875)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ True`

---

cov_expr

ARM BOOLEAN 100%

Was this arm ever SCANNED for the expression / abundance blocks? **True on 2,231 of 2,450 arms.** AUTHORED

values: `True (2,231) Â· False (219)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ True`

---

cov_famrole

ARM BOOLEAN 100%

Was this arm ever SCANNED for the family-role block? **True on 2,231 of 2,450 arms.** AUTHORED

values: `True (2,231) Â· False (219)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ True`

---

cov_field

ARM BOOLEAN 100%

Was this arm ever SCANNED for the focal-field correlation block? **True on 571 of 2,450 arms.** AUTHORED

values: `False (1,879) Â· True (571)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ True`

---

cov_focal

ARM BOOLEAN 100%

Was this arm ever SCANNED for the focal-locus CNV block? **True on 733 of 2,450 arms.** AUTHORED

values: `False (1,717) Â· True (733)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ True`

---

cov_fst

ARM BOOLEAN 100%

Was the family-share block computed for this arm at all â€” i.e. did it have the abundance input. AUTHORED

values: `False (1,965) Â· True (485)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ False`

---

cov_gctx

ARM BOOLEAN 100%

Was this arm ever SCANNED for the genomic-context block? **True on 714 of 2,450 arms.** AUTHORED

values: `False (1,736) Â· True (714)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ True`

---

cov_isocomp

ARM BOOLEAN 100%

Was this arm ever SCANNED for the isomiR-composition block? **True on 2,222 of 2,450 arms.** AUTHORED

values: `True (2,222) Â· False (228)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ True`

---

cov_isomir

ARM BOOLEAN 100%

Was this arm ever SCANNED for the isomiR seed-shift block? **True on 687 of 2,450 arms.** AUTHORED

values: `False (1,763) Â· True (687)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ True`

---

cov_meth

ARM BOOLEAN 100%

Was this arm ever SCANNED for the locus-methylation (`bc_meth_`) block? **True on 343 of 2,450 arms.** AUTHORED

values: `False (2,107) Â· True (343)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ True`

---

cov_real

ARM BOOLEAN 100%

Was this arm ever SCANNED for the realized-coupling (`real_`) block? **True on 590 of 2,450 arms.** AUTHORED

values: `False (1,860) Â· True (590)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ True`

---

cov_seed_scan
ARM BOOLEAN 100%

Was this arm ever SCANNED for the MANE 3'UTR seed scan (`site_`)? AUTHORED
values: True (2,370) Â· False (80)
example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

cov_seq_scan
ARM BOOLEAN 100%

Was this arm ever SCANNED for the scanMiR K_D scan (`seq_`)? AUTHORED
values: False (1,714) Â· True (736)
example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

cov_site_types
ARM BOOLEAN 100%

Was this arm ever SCANNED for the site-type breakdown? AUTHORED
values: False (1,732) Â· True (718)
example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

cov_subtype
ARM BOOLEAN 100%

Was this arm ever SCANNED for the PAM50 subtype block (`sub_`)? AUTHORED
values: False (1,722) Â· True (728)
example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

cov_targetscan
ARM BOOLEAN 100%

Was this arm ever SCANNED for the TargetScan context++ block (`ts_`)? AUTHORED
values: False (2,133) Â· True (317)
example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

SEQ_ Â· 4

reference â€" context, provenance and rarely-quoted detail

4 columns Â· rung **arm** Â· coverage **30%â€"30%** (median 30%)
2× count Â· 2× real â‰¥ 0

seq_n_genes_any
ARM COUNT AGG EDGE 30%

Genes with ANY scanMiR K_D site â€" AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1.124e+04

seq_n_genes_strong
ARM COUNT AGG EDGE 30%

Genes with a STRONG scanMiR K_D site â€" median 3,632, spearman **+0.822** with `sdsz_N_eff` and **+0.797** with `site_n_genes_canonical`, but only **+0.566** with `ts_n_genes`. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 4550

seq_promisc
ARM REAL Â‰¥ 0 30%

log-scale promiscuity over STRONG scanMiR sites (4.45 â€" 8.94). AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 8.423

seq_promisc_any
ARM REAL Â‰¥ 0 30%

Promiscuity over ANY scanMiR site (6.46 â€" 9.45) â€" the permissive cut. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 9.327

ISO_ Â· 5

reference â€" context, provenance and rarely-quoted detail

5 columns Â· rung **arm** Â· coverage **28%â€"28%** (median 28%)
2× fraction 0â€"1 Â· 1× count Â· 1× real â‰¥ 0 Â· 1× real, signed

iso_n_samples
ARM COUNT 28%

Samples behind the isomiR estimate. Range **3 â€" 1078**, 28% filled. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1078

> *Defined on: arms with a 5'-isomiR seed-shift measurement (687) â€" âš gate on iso_n_samples, the median is n-DEPENDENT*

iso_rpm_summed

ARM | REAL Â‰¥ 0 | 28%

âš **A SUM ACROSS SAMPLES, not a per-sample RPM â€” the name misleads.** Max **270,988,199**, which is impossible for a per-million unit; divided by `iso_n_samples` the scale is sane (median **57**, max 251,381). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 3.553e+07`

---

iso_seed_shift_frac

ARM | FRACTION 0â€“1 | 28%

Fraction of the arm's reads carrying a SHIFTED seed. Range **0 â€“ 0.868**, 28% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.004`

> *Defined on: arms with a 5'-isomiR seed-shift measurement (687) â€” âš gate on iso_n_samples, the median is n-DEPENDENT*

---

iso_seed_shift_sd

ARM | FRACTION 0â€“1 | 28%

Dispersion of that shift across samples. Range **0 â€“ 0.354**, 28% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.004`

> *Defined on: arms with a 5'-isomiR seed-shift measurement (687) â€” âš gate on iso_n_samples, the median is n-DEPENDENT*

---

iso_top_shift_nt

ARM | REAL, SIGNED | 28%

The most common shift, in nucleotides. Range **-6 â€“ 6**, 28% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 3`

> *Defined on: arms with a 5'-isomiR seed-shift measurement (687) â€” âš gate on iso_n_samples, the median is n-DEPENDENT*

CUR_HE_ Â· 2

reference â€” context, provenance and rarely-quoted detail

2 columns Â· rung **arm** Â· coverage **36%â€“36%** (median 36%)
2× count

cur_he_degree

ARM | COUNT | AGG EDGE | 36%

How many HE genes this arm regulates in the curated design â€” the arm's curation degree (median 3, max 125). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 51`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€” â›” FAME axes, not targetome (MH-208)*

---

cur_he_degree_expr

ARM | COUNT | AGG EDGE | 36%

As `cur_he_degree` but counting only targets above the expression floor. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 43`

FIELD_ Â· 5

reference â€” context, provenance and rarely-quoted detail

5 columns Â· rung **arm** Â· coverage **23%â€“23%** (median 23%)
4× signed âˆ'1â€¦1 Â· 1× count

field_excess_over_perm

ARM | SIGNED Âˆ'1Â€¦1 | 23%

â›”â›” **NOT A RETENTION RATIO â€” a FIFTH sense of the word in this codebase.** Measured: it equals **`field_r_own âˆ' field_r_perm`** (exact within rounding on 100% of 571 arms, max residual 0.001), i.e. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.363`

---

field_n

ARM | COUNT | 23%

Patients behind the field correlation. Range **40 â€“ 103**, 23% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 103`

> *Defined on: arms with a within-patient NAT field-effect fit (571)*

---

field_r_adj

ARM | SIGNED Âˆ'1Â€¦1 | 23%

The same, covariate-adjusted. Range **-0.344 â€“ +0.665**, 23% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.367`

> *Defined on: arms with a within-patient NAT field-effect fit (571)*

field_r_own
ARM SIGNED Â˘’1Â€¦1 23%

The arm's own field correlation (median +0.11 against a permutation null centred at 0.000). Range **-0.304 â€“ +0.624**, 23% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.343`

> *Defined on: arms with a within-patient NAT field-effect fit (571)*

---

field_r_perm
ARM SIGNED Â˘’1Â€¦1 23%

The permutation null for `field_r_own` â€” median **0.000**, range [â˘’0.11, +0.09]. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ -0.02`

BC_ Â· 11

reference â€” context, provenance and rarely-quoted detail

> **11** columns Â· rung **arm** Â· coverage **14%â€“36%** (median 14%)
> 4× fraction 0â€“1 Â· 4× count Â· 1× boolean Â· 1× signed â˘’1â€¦1 Â· 1× categorical

bc_fam_context_homogeneous
ARM BOOLEAN 36%

The arm's SEED-FAMILY property `fam_context_homogeneous`, broadcast down to the arm. AUTHORED

values: `True (720) Â· False (160)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ False`

---

bc_fam_dose_coding_host_frac
ARM FRACTION 0Â€“1 36%

The arm's SEED-FAMILY property `fam_dose_coding_host_frac`, broadcast down to the arm. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.121`

---

bc_fam_dose_cotranscribed_frac
ARM FRACTION 0Â€“1 36%

The arm's SEED-FAMILY property `fam_dose_cotranscribed_frac`, broadcast down to the arm. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.105`

---

bc_fam_n_distinct_coding_hosts
ARM COUNT 36%

The arm's SEED-FAMILY property `fam_n_distinct_coding_hosts`, broadcast down to the arm. AUTHORED

values: `1.0 (424) Â· 0.0 (379) Â· 2.0 (45) Â· 3.0 (32)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 3`

---

bc_fam_n_members
ARM COUNT 36%

The arm's SEED-FAMILY property `fam_n_members`, broadcast down to the arm. AUTHORED

values: `1.0 (606) Â· 2.0 (139) Â· 3.0 (69) Â· 5.0 (22) Â· 10.0 (21) Â· 8.0 (8) Â· 4.0 (8) Â· 6.0 (7)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 10`

---

bc_meth_dbeta
ARM SIGNED Â˘’1Â€¦1 14%

Tumour-minus-normal methylation beta at this arm's locus (median â˘’0.03, range â˘’0.32â€¦+0.23). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.1451`

---

bc_meth_direction
ARM CATEGORICAL 14%

Tumour-vs-normal methylation direction at this arm's locus. AUTHORED

values: `none (287) Â· hypo (46) Â· hyper (10)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ none`

---

bc_meth_n_cgi
ARM COUNT 14%

Locus methylation: `n_cgi`. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0`

---

bc_meth_n_probes
ARM COUNT 14%

Locus methylation: `n_probes`. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 1`

---

bc_meth_normal_beta
ARM FRACTION 0Â€“1 14%

Locus methylation: `normal_beta`. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.6494`

bc_meth_tumour_beta
ARM FRACTION 0â€“1 14%

Locus methylation: `tumour_beta`. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.7945

FAMROLE_ Â· 6

reference â€" context, provenance and rarely-quoted detail

6 columns Â· rung **arm** Â· coverage **88%â€“100%** (median 91%)
3× count Â· 1× fraction 0â€“1 Â· 1× categorical Â· 1× boolean

famrole_abund_rank
ARM COUNT 91%

The arm's rank by abundance within its seed family. Range **1 â€“ 14**, 91% filled. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1

> *Defined on: arms with a seed family â€" the arm's ROLE inside it; â›" DEGENERATE at famrole_n_members==1*

famrole_abund_share
ARM FRACTION 0â€“1 91%

Its share of the family's abundance â€" **median 1.000**, because most families are singletons. Range **0 â€“ 1**, 91% filled. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.465

> *Defined on: arms with a seed family â€" the arm's ROLE inside it; â›" DEGENERATE at famrole_n_members==1*

famrole_class
ARM CATEGORICAL 100%

The arm's role in its seed family â€" **`sole` on 1,639 of 2,450 (66.9%)**, `dominant` 282, `co` and the rest below. AUTHORED
values: sole (1,639) Â· dominant (282) Â· co (267) Â· minor (262)
example: hsa-let-7a-5p / let-7-5p/98-5p â†' co

famrole_is_dominant
ARM BOOLEAN 91%

Does the arm carry >50% of its family's abundance. Range **1,779 true / 452 false**, 91% filled. AUTHORED
values: True (1,779) Â· False (452)
example: hsa-let-7a-5p / let-7-5p/98-5p â†' False

> *Defined on: arms with a seed family â€" the arm's ROLE inside it; â›" DEGENERATE at famrole_n_members==1*

famrole_n_members
ARM COUNT 100%

How many arms are in the family. Range **1 â€“ 22**, 100% filled. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 10

> *Defined on: arms with a seed family â€" the arm's ROLE inside it; â›" DEGENERATE at famrole_n_members==1*

famrole_var_rank
ARM COUNT 88%

Its rank by variance within the family. Range **1 â€“ 14**, 88% filled. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 7

> *Defined on: arms with a seed family â€" the arm's ROLE inside it; â›" DEGENERATE at famrole_n_members==1*

GCTX_ Â· 16

reference â€" context, provenance and rarely-quoted detail

16 columns Â· rung **arm** Â· coverage **24%â€“29%** (median 29%)
6× categorical Â· 4× boolean Â· 3× identifier Â· 1× constant Â· 1× count Â· 1× real â‰¥ 0

gctx_clustered
ARM BOOLEAN 29%

Does this arm's precursor sit in a genomic CLUSTER with other miRNAs (310 true / 404 false)? AUTHORED
values: False (404) Â· True (310)
example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

gctx_context_mixed
ARM BOOLEAN 29%

Do this arm's multiple loci DISAGREE about their genomic context â€" **20 arms of 714**? AUTHORED

values: False (694) Â· True (20)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' False

---

gctx_coord_source
ARM CONSTANT 29%

Provenance of the genomic-context coordinates. AUTHORED

values: gencode (714)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' gencode

---

gctx_host
ARM IDENTIFIER 27%

The HOST GENE the precursor sits inside (356 distinct; `ENSG00000269842`, `MEG9`, `MEG3` most common). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' MIR100HG

---

gctx_host_region
ARM IDENTIFIER 27%

WHERE in the host the precursor sits (74 distinct; `intron 2` and `intron 1` most common, 61 arms each). An intronic miRNA is spliced from its host's pre-mRNA; an exonic one competes with it. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' exon 41

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€" 714)*

---

gctx_host_type
ARM CATEGORICAL 27%

The host's BIOTYPE â€" `protein_coding` 364 Â· `lncRNA` 303, plus 5 rarer levels. AUTHORED

values: protein_coding (364) Â· lncRNA (303) Â· transcribed_unprocessed_pseudogene (2) Â· TEC (1) Â· transcribed_unitary_pseudogene (1) Â· processed_pseudogene (1) Â· IG_V_gene (1)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' lncRNA

---

gctx_in_exon
ARM BOOLEAN 29%

Is the precursor inside a host EXON (218 true / 496 false) rather than an intron? AUTHORED

values: False (496) Â· True (218)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

gctx_mir_class
ARM CATEGORICAL 29%

The arm's genomic class â€" `sense_coding_host` (337) / `sense_lncRNA_host` / `antisense_overlap` / intergenic. Range **4 levels**, 29% filled. AUTHORED

values: sense_coding_host (337) Â· sense_lncRNA_host (296) Â· intergenic (41) Â· antisense_overlap (40)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' sense_lncRNA_host

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€" 714)*

---

gctx_mirgenedb
ARM BOOLEAN 29%

Is this arm in **MirGeneDB** â€" the curated high-confidence miRNA set (526 true / 188 false)? AUTHORED

values: True (526) Â· False (188)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

gctx_n_loci
ARM COUNT 29%

How many genomic loci the precursor maps to. Range **1 â€" 3**, 29% filled. AUTHORED

values: 1.0 (642) Â· 2.0 (62) Â· 3.0 (10)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 3

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€" 714)*

---

gctx_promoter_class
ARM CATEGORICAL 29%

Whose promoter drives it â€" `host_shared` (346) / `own` (239) / shared. Range **4 levels**, 29% filled. AUTHORED

values: host_shared (346) Â· own (239) Â· shared_other (128) Â· unknown (1)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' host_shared

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€" 714)*

---

gctx_promoter_coord_class
ARM CATEGORICAL 29%

The same call from coordinates rather than annotation. Range **4 levels**, 29% filled. AUTHORED

values: host_shared (480) Â· shared_other (111) Â· unknown (85) Â· intergenic (38)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' host_shared

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€" 714)*

gctx_promoter_gene
ARM IDENTIFIER 24%

The gene whose promoter drives the precursor. Range **307 distinct**, 24% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' MIR100HG`

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€" 714)*

---

gctx_promoter_gene_type
ARM CATEGORICAL 24%

That gene's biotype. Range **5 levels**, 24% filled. AUTHORED

values: protein_coding (320) Â· lncRNA (267) Â· misc_RNA (2) Â· transcribed_processed_pseudogene (1) Â· IG_V_gene (1)

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' lncRNA`

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€" 714)*

---

gctx_promoter_host_tss_kb
ARM REAL Â‰¥ 0 24%

Distance in kb from the precursor to its host's TSS (median **0.5 kb**, max 874). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 376.1`

---

gctx_strand_rel
ARM CATEGORICAL 27%

Is the precursor SENSE (633 arms) or ANTISENSE (40) to its host? AUTHORED

values: sense (633) Â· antisense (40)

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' sense`

SURV_ Â· 8

reference â€" context, provenance and rarely-quoted detail

> **8** columns Â· rung **arm** Â· coverage **6%â€"6%** (median 6%)
> 4× p-value Â· 3× real â‰¥ 0 Â· 1× constant

surv_dfi_hr_adj
ARM REAL Â‰¥ 0 6%

Adjusted DFI hazard ratio per arm â€" range **[0.84, 1.16]**, median **1.00**. The whole distribution sits within Â±16% of no effect. See `surv_dfi_q_adj`. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.9371`

> *Defined on: arms in the TCGA outcome panel (145) â€" â›" TCGA legs only; the GSE leg is bare-stem matched*

---

surv_dfi_hr_base
ARM REAL Â‰¥ 0 6%

Unadjusted DFI hazard ratio for the arm (0.86 â€" 1.16, median 1.00). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.9463`

---

surv_dfi_p_adj
ARM P-VALUE 6%

Adjusted DFI p. Range **0.0741 â€" 0.997**, 6% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.4701`

> *Defined on: arms in the TCGA outcome panel (145) â€" â›" TCGA legs only; the GSE leg is bare-stem matched*

---

surv_dfi_p_base
ARM P-VALUE 6%

Unadjusted DFI p for this arm. Range **0.0973 â€" 0.998**, 6% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.5368`

> *Defined on: arms in the TCGA outcome panel (145) â€" â›" TCGA legs only; the GSE leg is bare-stem matched*

---

surv_dfi_q_adj
ARM CONSTANT 6%

BH q for the arm's DFI hazard ratio. AUTHORED

values: 0.996528491082692 (145)

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.9965`

---

surv_os_hr_adj
ARM REAL Â‰¥ 0 6%

Adjusted OS hazard ratio. Range **0.816 â€" 1.24**, 6% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 1.011`

> *Defined on: arms in the TCGA outcome panel (145) â€" â›" TCGA legs only; the GSE leg is bare-stem matched*

---

surv_os_p_adj
ARM P-VALUE 6%

Adjusted OS p. Range **0.00235 â€" 0.998**, 6% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.8856`

> *Defined on: arms in the TCGA outcome panel (145) â€" â›" TCGA legs only; the GSE leg is bare-stem matched*

surv_os_q_adj

ARM | P-VALUE | 6%

BH q for OS â€” AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.9745`

AID_ Â· 4

reference â€" context, provenance and rarely-quoted detail

---

4 columns Â· rung **arm** Â· coverage **6%â€"6%** (median 6%)
1× fraction 0â€"1 Â· 1× real â‰¥ 0 Â· 1× signed âˆ'1â€¦1 Â· 1× count

aid_frac_resolvable

ARM | FRACTION 0â€"1 | AGG ARM-IN-FAMILY | 6%

Fraction of the arm's multi-arm cells that resolve â€" **median 0.19**. Range **0 â€" 1**, 6% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.4118`

> *Defined on: arms inside â‰¥1 MULTI-ARM (gene,family) cell â€" the arm rung's 20.3%-of-edges footprint (142)*

---

aid_med_arm_sep_z

ARM | REAL Â‰¥ 0 | AGG ARM-IN-FAMILY | 6%

Median separation z across those cells. Range **0.00108 â€" 5.59**, 6% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 2.381`

> *Defined on: arms inside â‰¥1 MULTI-ARM (gene,family) cell â€" the arm rung's 20.3%-of-edges footprint (142)*

---

aid_med_oof_drho

ARM | SIGNED Âˆ'1Â€¦1 | AGG ARM-IN-FAMILY | 6%

Median out-of-fold gain from resolving the arm. Range **-0.092 â€" +0.0421**, 6% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' -0.004584`

> *Defined on: arms inside â‰¥1 MULTI-ARM (gene,family) cell â€" the arm rung's 20.3%-of-edges footprint (142)*

---

aid_n_cells

ARM | COUNT | AGG ARM-IN-FAMILY | 6%

Multi-arm cells this arm participates in â€" fill **5.8%** (the arm identifiability block is the sparsest on the card). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 17`

WSHIFT_ Â· 6

reference â€" context, provenance and rarely-quoted detail

---

6 columns Â· rung **arm** Â· coverage **45%â€"91%** (median 52%)
3× real â‰¥ 0 Â· 2× real, signed Â· 1× count

wshift_mean_dGlobalRank

ARM | REAL, SIGNED | 52%

Mean within-patient change in its global rank. Range **-36.2 â€" 42.7**, 52% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' -0.13`

> *Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)*

---

wshift_mean_own_shift

ARM | REAL, SIGNED | 52%

Mean within-patient change in the arm's own share. Range **-3.94 â€" 5.05**, 52% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' -0.223`

> *Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)*

---

wshift_n_pairs

ARM | COUNT | 91%

Paired patients behind the within-patient shift. Range **0 â€" 103**, 91% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 103`

> *Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)*

wshift_nat_patient_sd
ARM | REAL Â‰¥ 0 | 74%

Between-patient SD of the arm's NAT level. Range **0 â€“ 2.73**, 74% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.556`

> *Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)*

---

wshift_own_specific_frac
ARM | REAL Â‰¥ 0 | 45%

Within-patient paired quantities on the ~103 matched tumour/NAT participants. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.62`

---

wshift_sd_dGlobalRank
ARM | REAL Â‰¥ 0 | 45%

Dispersion of that change. Range **0.03 â€“ 50.9**, 45% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.4`

> *Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)*

CNV_ Â· 9

reference â€” context, provenance and rarely-quoted detail

9 columns Â· rung **arm** Â· coverage **35%â€“35%** (median 35%)
6× percent Â· 3× real â‰¥ 0

cnv_mean_cn
ARM | REAL Â‰¥ 0 | 35%

Arm-locus copy number: MEAN copy number at this arm's locus, over the TCGA cohort. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 2.403`

---

cnv_median_cn
ARM | REAL Â‰¥ 0 | 35%

Arm-locus copy number: MEDIAN copy number, over the TCGA cohort. AUTHORED

values: 2.0 (632) Â· 3.0 (175) Â· 4.0 (40) Â· 2.5 (8) Â· 3.5 (1)

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 2`

---

cnv_pct_altered
ARM | PERCENT | 35%

Arm-locus copy number: percent of patients with ANY alteration, over the TCGA cohort. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 43.66`

---

cnv_pct_amp
ARM | PERCENT | 35%

Arm-locus copy number: percent AMPLIFIED, over the TCGA cohort. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 16.21`

---

cnv_pct_deep_del
ARM | PERCENT | 35%

Percent of samples with a deep deletion at this arm's locus. AUTHORED

values: 0.0 (722) Â· 0.0953 (75) Â· 0.1907 (32) Â· 0.286 (10) Â· 0.0954 (6) Â· 0.3813 (6) Â· 1.5253 (4) Â· 0.0955 (1)

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0`

---

cnv_pct_gain
ARM | PERCENT | 35%

Arm-locus copy number: percent with a low-level GAIN, over the TCGA cohort. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 16.11`

---

cnv_pct_loh
ARM | PERCENT | 35%

Arm-locus copy number: percent with loss of heterozygosity, over the TCGA cohort. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 69.97`

---

cnv_pct_loss
ARM | PERCENT | 35%

Arm-locus copy number: percent with a LOSS, over the TCGA cohort. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 11.34`

---

cnv_sd_cn
ARM | REAL Â‰¥ 0 | 35%

Arm-locus copy number: SD of copy number across patients, over the TCGA cohort. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.9746`

TIER_ Â· 5

reference â€” context, provenance and rarely-quoted detail

5 columns Â· rung **arm** Â· coverage **91%â€“91%** (median 91%)
2× real â‰¥ 0 Â· 1× categorical Â· 1× boolean Â· 1× fraction 0â€“1

tier_class
ARM CATEGORICAL 91%

Expression tier â€” **silent on 1,518 of 2,236 arms (67.9%)**, conditional 476, the rest robust. AUTHORED

values: silent (1,518) Â· conditional (476) Â· robust (242)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' robust

tier_expressed
ARM BOOLEAN 91%

Is the arm above the expression floor often enough to be usable (718 true / 1,518 false)? AUTHORED

values: False (1,518) Â· True (718)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

tier_frac_expressed
ARM FRACTION 0â€“1 91%

Fraction of samples where it is expressed â€” **median 0.00**. Range **0 â€“ 1**, 91% filled. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1

> *Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)*

tier_max
ARM REAL â‰¥ 0 91%

The arm's maximum RPM across tumour samples. Range **0.105 â€“ 19.4**, 91% filled. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 17.48

> *Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)*

tier_p90
ARM REAL â‰¥ 0 91%

The arm's 90th-percentile RPM across tumour samples (0.10 â€“ 18.49, median 0.94). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 15.76

FOC_ Â· 7

reference â€” context, provenance and rarely-quoted detail

7 columns Â· rung **arm** Â· coverage **30%â€“30%** (median 30%)
2× count Â· 2× signed âˆ'1â€¦1 Â· 1× identifier Â· 1× boolean Â· 1× p-value

foc_focal_locus
ARM IDENTIFIER 30%

The focal MIRBase locus used for this arm's CN analysis (437 distinct). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' MI0000062

foc_multilocus
ARM BOOLEAN 30%

Does this arm map to MORE THAN ONE genomic locus (35 true / 698 false)? AUTHORED

values: False (698) Â· True (35)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

foc_n
ARM COUNT 30%

Patients behind the focal-locus CN correlation. Range **41 â€“ 1032**, 30% filled. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1032

> *Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€” â›” DESCRIPTIVE, not an instrument*

foc_n_loci
ARM COUNT 30%

How many loci the arm maps to. Range **1 â€“ 3**, 30% filled. AUTHORED

values: 1.0 (698) Â· 2.0 (27) Â· 3.0 (8)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 3

> *Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€” â›” DESCRIPTIVE, not an instrument*

foc_partial_p

ARM · P-VALUE · 30%

p for that partial correlation. Range **1.54e-42 â€“ 0.996**, 30% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 6.541e-09`

> *Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€" â›" DESCRIPTIVE, not an instrument*

---

foc_partial_rho

ARM · SIGNED âˆ'1â€¦1 · 30%

The same, conditioned on covariates. Range **-0.269 â€“ +0.424**, 30% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.1871`

> *Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€" â›" DESCRIPTIVE, not an instrument*

---

foc_spearman_rho

ARM · SIGNED âˆ'1â€¦1 · 30%

Correlation between the arm's dose and its focal locus CN. Range **-0.235 â€“ +0.485**, 30% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.08216`

> *Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€" â›" DESCRIPTIVE, not an instrument*

SURROGATE_ Â· 2

reference â€" context, provenance and rarely-quoted detail

2 columns Â· rung **arm** Â· coverage **0%â€“0%** (median 0%)
1× fraction 0â€“1 Â· 1× categorical

surrogate_corr

ARM · FRACTION 0â€“1 · 0%

Correlation between this arm and its surrogate (0.62 â€“ 0.89) â€" the surrogate's quality gate. See the edge twin. AUTHORED

values: `0.621 (1) Â· 0.874 (1) Â· 0.892 (1) Â· 0.842 (1)`

example: `hsa-miR-181a-5p / miR-181-5p â†' 0.892`

> *Defined on: arms with a same-seed SURROGATE (MH-166) â€" 3.6% of edges*

---

surrogate_instrument

ARM · CATEGORICAL · 0%

Which surrogate stands in for this arm's collapsed GTEx leg (4 levels). AUTHORED

values: `hsa-miR-93-5p (1) Â· hsa-miR-20a-5p (1) Â· hsa-miR-181b-5p (1) Â· hsa-miR-429 (1)`

example: `hsa-miR-181a-5p / miR-181-5p â†' hsa-miR-181b-5p`

CNVC_ Â· 5

reference â€" context, provenance and rarely-quoted detail

5 columns Â· rung **arm** Â· coverage **31%â€“31%** (median 31%)
2× p-value Â· 2× signed âˆ'1â€¦1 Â· 1× count

cnvc_n

ARM · COUNT · 31%

Patients behind the CN-coupling estimate. Range **20 â€“ 1032**, 31% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 1032`

> *Defined on: arms with a CNâ†"expression concordance fit (756)*

---

cnvc_partial_p

ARM · P-VALUE · 31%

p for the conditioned CN correlation. Range **1.25e-42 â€“ 1**, 31% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.0001062`

> *Defined on: arms with a CNâ†"expression concordance fit (756)*

---

cnvc_partial_q

ARM · P-VALUE · 31%

BH q for the arm's CN-coupling partial correlation. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.0006635`

---

cnvc_partial_rho

ARM · SIGNED âˆ'1â€¦1 · 31%

The same, conditioned â€" barely moves (+0.05 vs +0.06), so the CN association is not carried by the covariates. Range **-0.322 â€“ +0.468**, 31% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.1238`

> *Defined on: arms with a CNâ†"expression concordance fit (756)*

cnvc_spearman_rho
ARM · SIGNED Âˆ'1Â€¦1 · 31%

Arm dose vs locus CN, unconditioned. Range **-0.378 â€“ +0.5**, 31% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.0573`

> *Defined on: arms with a CNâ†”expression concordance fit (756)*

DISC_ Â· 2

reference â€” context, provenance and rarely-quoted detail

---

2 columns Â· rung **arm** Â· coverage **1%â€“100%** (median 50%)
1× categorical Â· 1× count

disc_gold_families
ARM · CATEGORICAL · 1%

Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed). AUTHORED

example: `hsa-miR-25-3p / miR-25-3p/32-5p/92-3p/363-3p/367-3p â†' miR-25-3p/32-5p/92-3p/363-3p/367-3p`

---

disc_n_gold_edges
ARM · COUNT · 100%

How many of the 157 discovery-queue edges name this arm. Concentrated: miR-106b-5p **39** Â· miR-20a-5p 25 Â· miR-93-5p 25 Â· miR-19a-3p 19 Â· miR-18a-5p 9. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0`

CLOAD_ Â· 3

reference â€” context, provenance and rarely-quoted detail

---

3 columns Â· rung **arm** Â· coverage **8%â€“60%** (median 60%)
1× real â‰¥ 0 Â· 1× fraction 0â€“1 Â· 1× categorical

cload_dose_rpm
ARM · REAL Â‰¥ 0 · 60%

The arm's dose used in the compartment-loading fit. Range **0.279 â€“ 2.4e+05**, 60% filled. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 3.285e+04`

> *Defined on: arms with a compartment-loading row (1,473) â€” âš âš its source has NO PRODUCER (MH-111 ad-hoc)*

---

cload_lock
ARM · FRACTION 0Â€“1 · 8%

Is this arm's compartment loading LOCKED to one cell type rather than spread across the deconvolution? AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.256`

---

cload_top_compartment
ARM · CATEGORICAL · 60%

Which compartment the arm's abundance loads on most. Range **8 levels**, 60% filled. AUTHORED

values: `CAFs (1,335) Â· Endothelial (47) Â· PVL (35) Â· Myeloid (31) Â· Normal Epithelial (15) Â· B-cells (4) Â· Plasmablasts (3) Â· T-cells (3)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' Endothelial`

> *Defined on: arms with a compartment-loading row (1,473) â€” âš âš its source has NO PRODUCER (MH-111 ad-hoc)*

MODEL_ Â· 1

reference â€” context, provenance and rarely-quoted detail

---

1 columns Â· rung **arm** Â· coverage **30%â€“30%** (median 30%)
1× count

model_n_genes
ARM · COUNT · AGG EDGE · 30%

Which model variant produced this row. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 38`

> *Defined on: arms carrying >=1 curated HE edge in the model design (741)*

COMV_ Â· 2

reference â€" context, provenance and rarely-quoted detail

2 columns Â· rung **arm** Â· coverage **15%â€"15%** (median 15%)
1× count Â· 1× identifier

comv_module
ARM COUNT 15%

Which co-movement module this arm belongs to in the Â§6b-RETIRED heuristic lane. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 22`

comv_top_partner
ARM IDENTIFIER 15%

Co-expression module membership and top co-expressed partner. Pure co-expression â€" the cross-state class columns are deliberately excluded from this block. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' hsa-let-7f-5p(0.67);hsa-let-7b-5p(0.55);hsa-let-7e-5p(0.43);hsa-let-7b-3p(0.40);hsa-miR-101-3p.1(0.36)`

> *Defined on: arms in a co-expression module (371) â€" pure co-expression; the state-class columns are excluded*

HX_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **arm** Â· coverage **28%â€"28%** (median 28%)
1× real â‰¥ 0

hx_hly_baseline_imputed
ARM REAL Â‰¥ 0 28%

â›" IMPUTED healthy baseline (miTED rank-transfer or seed-mate substitution) â€" LABELLED as imputed, never presented as a measurement. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 12.32`

seed_family_card.tsv

seed_family

One row per the seed family itself, gene-free. 83.7% of families are SINGLETONS â€” for those every 'family-level' statistic is trivially that one arm's value. Mask on `fam_single_member` first.

33 columns **1,959** rows **2** blocks median coverage **92%**

Contents â€” 2 blocks, most important first

SEED_ Â· 1

core â€” the columns findings rest on

1 columns Â· rung **key** Â· coverage **100%â€“100%** (median 100%)
1× identifier

seed_family
KEY IDENTIFIER 100%

KEY â€” the seed family itself, one row per family. AUTHORED
example: miR-4670-3p

FAM_ Â· 32

reference â€” context, provenance and rarely-quoted detail

32 columns Â· rung **family** Â· coverage **4%â€“100%** (median 91%)
12× count Â· 8× real â‰¥ 0 Â· 7× fraction 0â€“1 Â· 2× boolean Â· 1× categorical Â· 1× identifier Â· 1× signed âˆ’1â€¦1

fam_ctx_composition
FAMILY CATEGORICAL 29%

The family's genomic-context MIX as a tally string â€” `sense_coding_host:1` on 251 families, `sense_lncRNA_host:1` on 175, `antisense_overlap:1` next (29 distinct patterns). AUTHORED
example: miR-34b-5p/449c-5p â†’ sense_lncRNA_host:1,sense_coding_host:1

fam_ctx_context_homogeneous
FAMILY BOOLEAN 29%

Do ALL of this family's members share one genomic context â€” **True on 524 of 571 (91.8%)**? AUTHORED
values: True (524) Â· False (47)
example: miR-34b-5p/449c-5p â†’ False

fam_ctx_dose_coding_host_frac
FAMILY FRACTION 0â€“1 29%

Fraction of the family's DOSE that comes from members with a protein-coding host. AUTHORED
example: miR-34b-5p/449c-5p â†’ 0

fam_ctx_dose_cotranscribed_frac
FAMILY FRACTION 0â€“1 29%

Fraction of the family's dose from members CO-TRANSCRIBED with their host (sense strand). AUTHORED
example: miR-34b-5p/449c-5p â†’ 0

fam_ctx_n_distinct_coding_hosts
FAMILY COUNT 29%

How many DISTINCT coding hosts the family's members sit in. AUTHORED
values: `1.0 (288) Â· 0.0 (264) Â· 2.0 (17) Â· 3.0 (2)`
example: `miR-34b-5p/449c-5p â†' 1`

---

fam_cur_degree_max
FAMILY COUNT 36%

The largest curated HE-degree among the family's members. AUTHORED
example: `miR-3682-3p â†' 1`

---

fam_dominant_arm
FAMILY IDENTIFIER 92%

WHICH member carries the largest share of the family's dose. AUTHORED
example: `miR-4691-5p/6792-3p â†' hsa-miR-4691-5p`

---

fam_dominant_share
FAMILY FRACTION 0â€“1 92%

The top member's share of the family's total dose. AUTHORED
example: `miR-4691-5p/6792-3p â†' 0.5101`

---

fam_dose_hhi
FAMILY FRACTION 0â€“1 15%

Concentration of DOSE across the family's members. âœ... **FLOOR-CORRECTED â€" verified 2026-08-19**: its minimum sits far below the raw 1/k floor at every member count (0.0000 at k=2 where the raw floor is 0.500), and spearman(n_members, hhi) = **+0.1896** rather than the strongly negative value a raw index would give. AUTHORED
example: `miR-378-3p â†' 0.9346`

---

fam_dose_max_log2
FAMILY REAL â‰¥ 0 92%

The largest member dose, log2. AUTHORED
example: `miR-4691-5p/6792-3p â†' 0.4237`

---

fam_dose_med_log2
FAMILY REAL â‰¥ 0 92%

The median member dose, log2. AUTHORED
example: `miR-4691-5p/6792-3p â†' 0.4164`

---

fam_dose_total_log2
FAMILY REAL â‰¥ 0 100%

The family's TOTAL dose, log2 â€" the denominator behind every dose share. AUTHORED
example: `miR-4670-3p â†' 0.3658`

---

fam_fame_npmid_sum
FAMILY COUNT 100%

Summed publication count over the family's members. AUTHORED
example: `miR-4670-3p â†' 13`

---

fam_n_expressed
FAMILY COUNT 100%

How many members clear the expression floor. AUTHORED
example: `miR-4670-3p â†' 1`

---

fam_n_guide
FAMILY COUNT 100%

How many members are classed GUIDE strand by AGO loading (0 â€“ 9). AUTHORED
values: `0.0 (1,737) Â· 1.0 (188) Â· 2.0 (23) Â· 3.0 (3) Â· 6.0 (3) Â· 5.0 (2) Â· 4.0 (2) Â· 9.0 (1)`
example: `miR-4670-3p â†' 0`

---

fam_n_members
FAMILY COUNT 100%

How many arms are in this seed family. AUTHORED
example: `miR-4670-3p â†' 1`

---

fam_n_members_context
FAMILY COUNT 29%

Arms in the seed family (median 1, max 22). AUTHORED
values: `1.0 (486) Â· 2.0 (61) Â· 3.0 (14) Â· 5.0 (4) Â· 10.0 (2) Â· 4.0 (2) Â· 6.0 (1) Â· 8.0 (1)`
example: `miR-34b-5p/449c-5p â†' 2`

---

fam_n_model_edges
FAMILY COUNT 100%

How many edges the family holds in the LEARNED MODEL's design (0 â€“ 209). AUTHORED
example: `miR-4670-3p â†' 0`

---

fam_n_passenger
FAMILY COUNT 100%

How many members are classed PASSENGER strand (0 â€“ 3). See `fam_n_guide`. AUTHORED
values: `0.0 (1,784) Â· 1.0 (167) Â· 2.0 (5) Â· 3.0 (3)`
example: `miR-4670-3p â†' 0`

> *Defined on: seed families in the arm-card universe (1,959; â‰ˆ 83.7% single-member â‡' shares/HHI are 1 or NaN)*

fam_n_shift_gt20_gated
FAMILY COUNT 100%

How many members show an isomiR seed shift above 20%, among those with enough coverage to judge. Range 0 â€“ 2, 100% filled â€” so this is a rare event, not a distribution. AUTHORED

values: `0.0 (1,911) Â· 1.0 (45) Â· 2.0 (3)`

example: `miR-4670-3p â†’ 0`

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_real_n_c10_sum
FAMILY COUNT 100%

Summed count of realized couplings below âˆ’0.10 over the family's members. AUTHORED

example: `miR-4670-3p â†’ 0`

---

fam_real_n_scored_sum
FAMILY COUNT 100%

Summed count of scored gene-regulator pairs over the family's members â€” the DENOMINATOR for `fam_real_n_c10_sum`. Range 0 â€“ 169, 100% filled. AUTHORED

example: `miR-4670-3p â†’ 0`

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_redund_m_eff
FAMILY REAL â‰¥ 0 4%

EFFECTIVE number of independent members after accounting for within-family correlation â€” the family's redundancy-corrected size (mode **2.0**, 50 distinct values). AUTHORED

example: `miR-25-3p/32-5p/92-3p/363-3p/367-3p â†’ 4.82`

---

fam_redund_median_within_rho
FAMILY SIGNED âˆ’1â€¦1 4%

Median pairwise correlation BETWEEN the family's members (âˆ’0.236 â€“ +1.00). AUTHORED

example: `miR-25-3p/32-5p/92-3p/363-3p/367-3p â†’ 0.178`

---

fam_redund_overcount_frac
FAMILY FRACTION 0â€“1 4%

How much the family's naive member count OVERSTATES its independent size, once within-family correlation is accounted for (0 â€“ 0.374). AUTHORED

example: `miR-25-3p/32-5p/92-3p/363-3p/367-3p â†’ 0.035`

---

fam_seed_shift_max_gated
FAMILY FRACTION 0â€“1 13%

The same, gated to well-covered members. AUTHORED

example: `miR-27-3p â†’ 0.023`

---

fam_single_member
FAMILY BOOLEAN 100%

ââ **THE DOMINANT FACT ABOUT THIS ENTIRE CARD: 1,639 of 1,959 families (83.7%) are SINGLETONS.** For those rows every 'family-level' statistic is TRIVIALLY that one arm's value, and `fam_dominant_share` is forced to exactly 1.0. AUTHORED

values: `True (1,639) Â· False (320)`

example: `miR-4670-3p â†’ True`

---

fam_targetome_seed_med
FAMILY REAL â‰¥ 0 97%

Median MANE 3'UTR seed-scan targetome over the members. AUTHORED

example: `miR-4688/6743-5p â†’ 1888`

---

fam_targetome_seq_med
FAMILY REAL â‰¥ 0 30%

Median scanMiR K_D targetome over the members. AUTHORED

example: `miR-3615 â†’ 2353`

---

fam_var_hhi
FAMILY FRACTION 0â€“1 14%

Herfindahl concentration of `abund_sd` ACROSS the family's members â€” is the family's variance carried by one arm? AUTHORED

example: `miR-376b-5p/376c-5p â†’ 6.327e-05`

---

fam_var_max
FAMILY REAL â‰¥ 0 89%

LARGEST member dispersion in the family. AUTHORED

example: `miR-4687-3p â†’ 0.2317`

---

fam_var_med
FAMILY REAL â‰¥ 0 89%

MEDIAN dispersion (`abund_sd`) over the family's members. AUTHORED

example: `miR-4687-3p â†’ 0.2317`